

Single particle tracking

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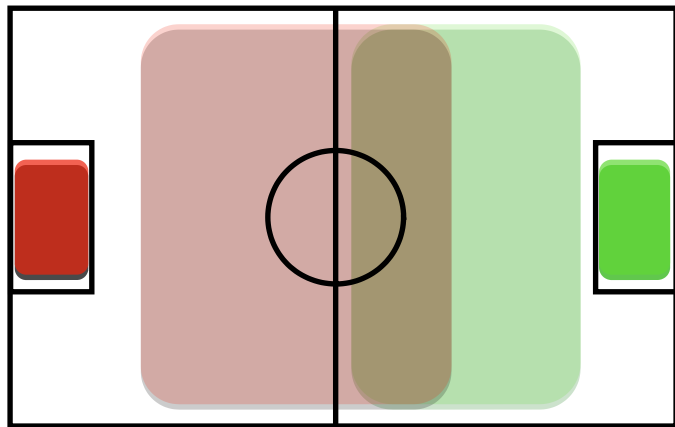
legall@cbs.cnrs.fr
<http://marcnol.weebly.com/>
<http://www.cbs.cnrs.fr/index.php/fr/>

2nd of December 2020

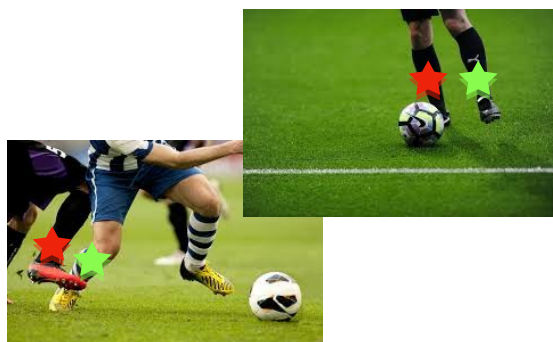
Soccer under the microscope



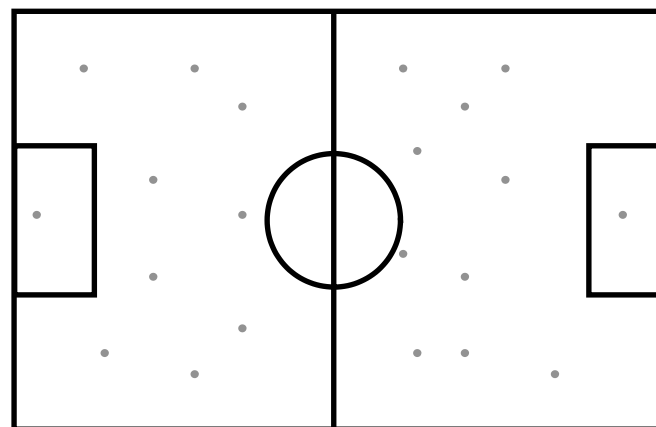
Diffraction-limited



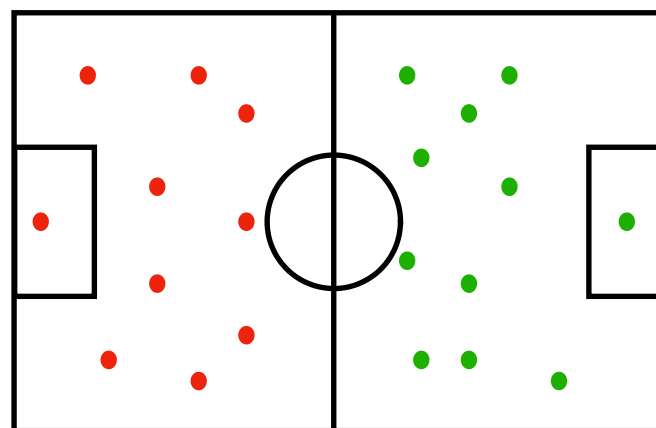
FRET



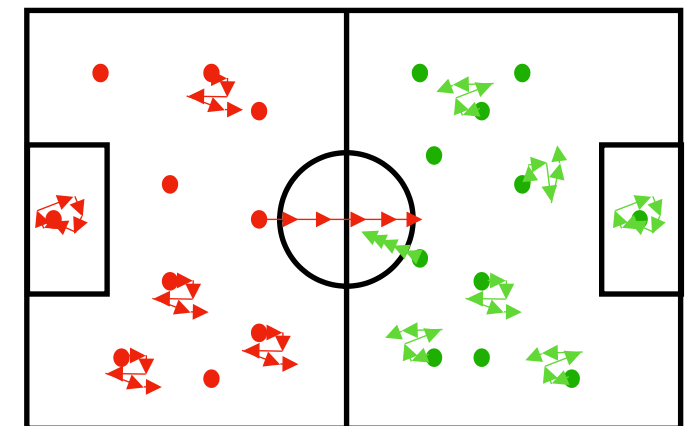
AFM/Cryo-EM



Super-resolution



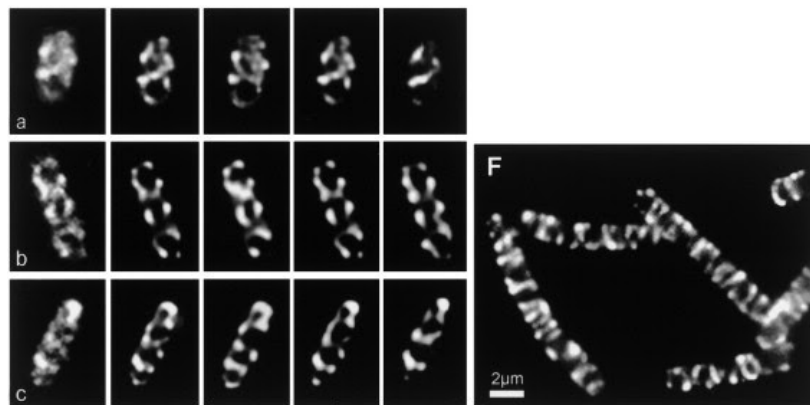
Single Particle Tracking



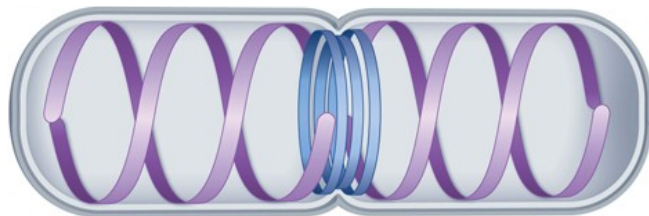
Why time information matters?

Mreb: bacterial actin-like proteins that move around the inside of the cell wall inserting and removing material to allow the cell to grow.

Single shot image



Jones et al, Control of Cell Shape in Bacteria: Helical, Actin-like Filaments in Bacillus subtilis, Cell, 2001



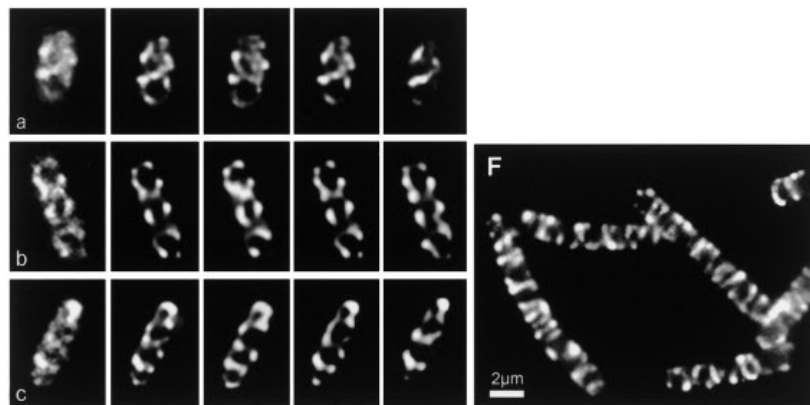
Juarez et al, A bacterial actin unites to divide bacterial cells, Emboj, 2012

**Long spiral filament
twisting along the cell
length**

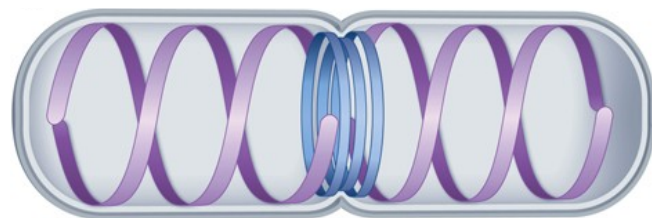
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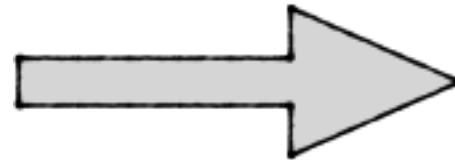
Single shot image



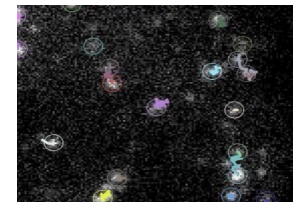
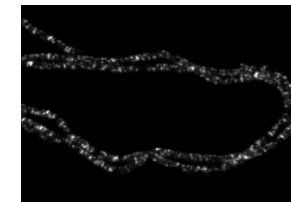
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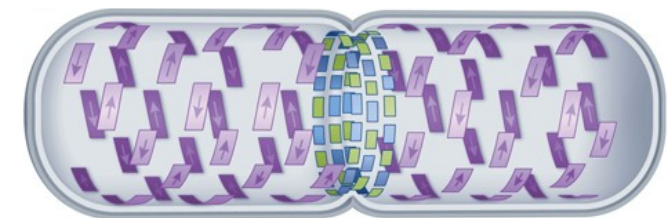
Higher temporal resolution



Movie



Garner lab : garnerlab.fas.harvard.edu/wordpress/



Juarez et al, A bacterial actin unites to divide bacterial cells, *Emboj*, 2012

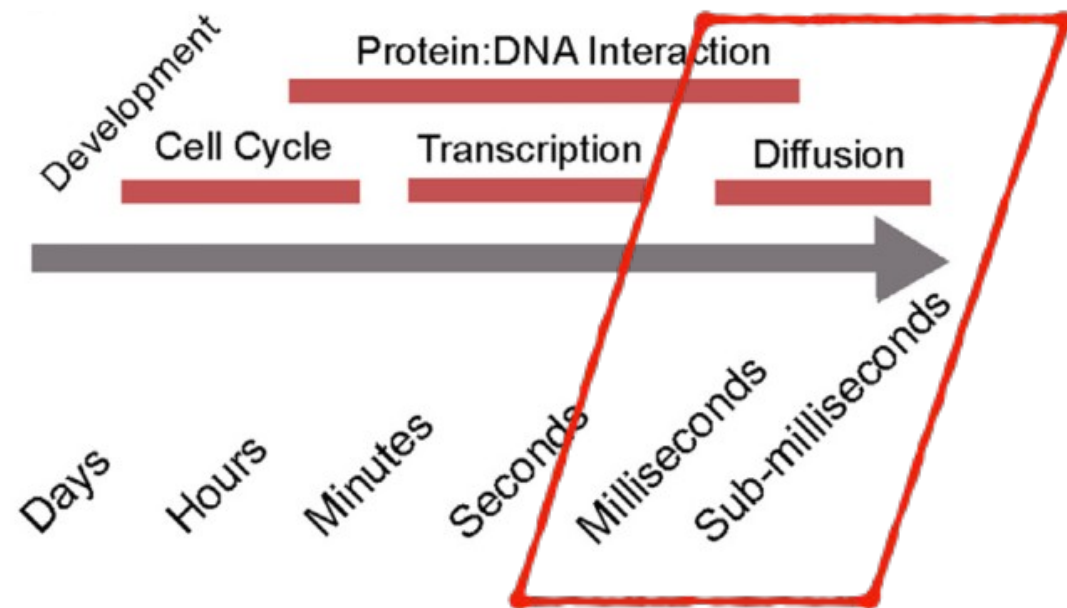
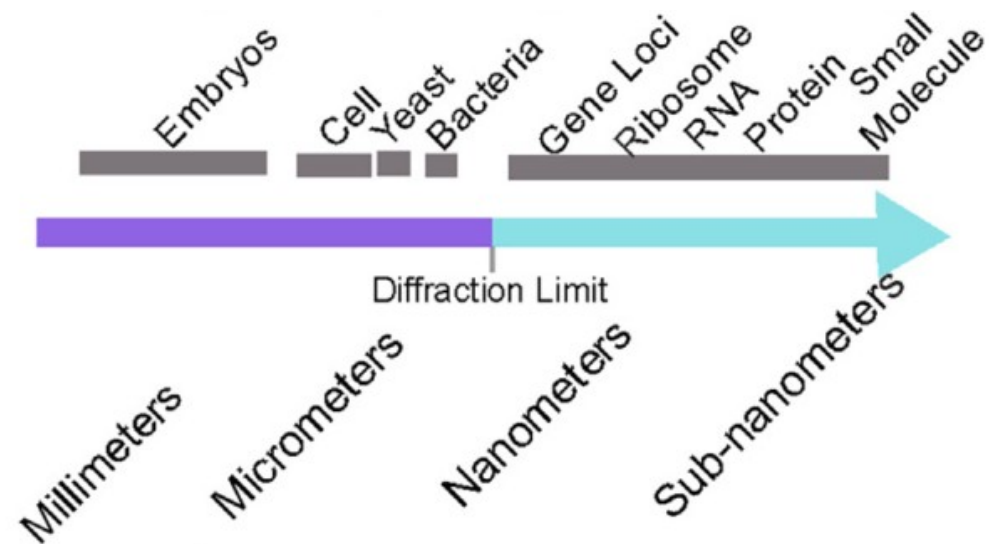
~~Long spiral filament
twisting along the cell
length~~

Short dynamic
patches of
polymers

Outline

- 1.** Introduction on Brownian motion
- 2.** Probes for single particle tracking
- 3.** Illumination for single particle tracking
- 4.** Analysis
- 5.** Going 3D
- 6.** Applications

Spatiotemporal length scales in biology



At the microscopic scale, particles move about in response to bombardment by the surrounding molecules.

Diffusion: moving molecules self-propelled by thermal energy

Are your acquisition parameters / approaches compatible with your biological question?

Introduction

Brownian motion

Why is diffusion a key parameter in biology?

FRAP/FCS for diffusion coefficients determination

Why single particle/molecule tracking?

Brownian motion / random walk

1827 : Robert Brown observed the random motion of micro-particles suspended in gases and liquids. He called this random or jiggling motion “Brownian motion”

1855 : Adolph Fick proposes a law in which the flux of diffusion is proportional to the gradient of concentration

1905 : Albert Einstein first explained this phenomenon on the basis of kinetic theory of molecules

1926 : Jean Perrin won the Nobel Prize for his work confirming Einstein’s hypothesis.

Although diffusion is non-directional, it can be an efficient transport mechanism in cells, particularly over short distances.



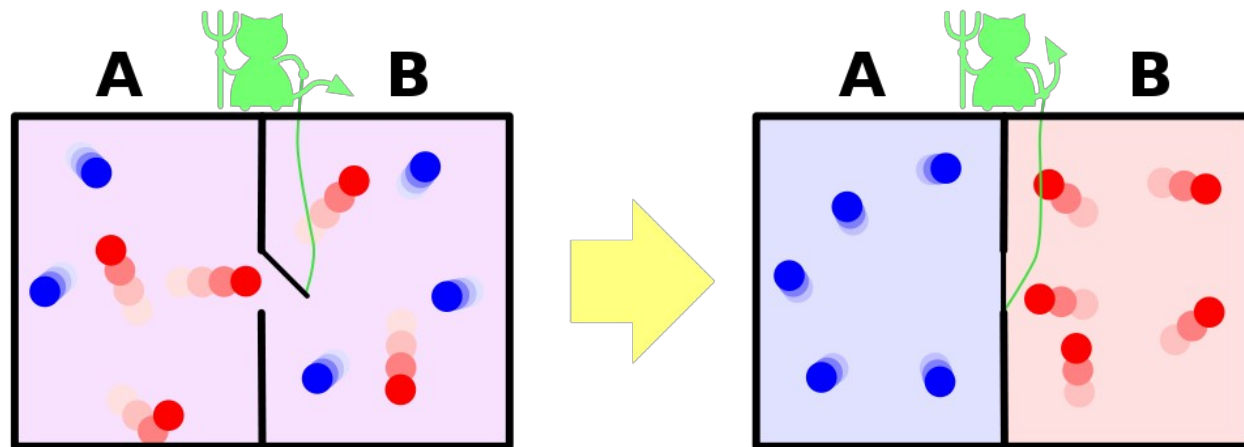
Experiment : Ink drop will spread faster in hot water (youtube).

Maxwell's demon

Second law of thermodynamics : Any isolated system will naturally tend to degenerate into a more disordered state. Entropy won't decrease.

Brownian motion $\Rightarrow$ homogeneity

Maxwell's experimental thought : *A demon opens and shuts the door so that fast molecules (red) pass into the other chamber, while slow molecules (blue) remain in the first chamber*



Because faster molecules are hotter, the demon's behavior causes one chamber to warm up as the other cools, thus decreasing entropy and violating the second law of thermodynamics.

**Brownian motion (+ demon)
 $\Rightarrow$ order**

Maxwell's demon in biological systems

**Membrane
microdomains**

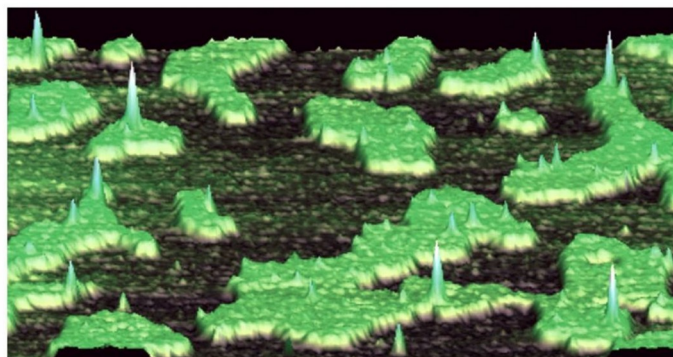
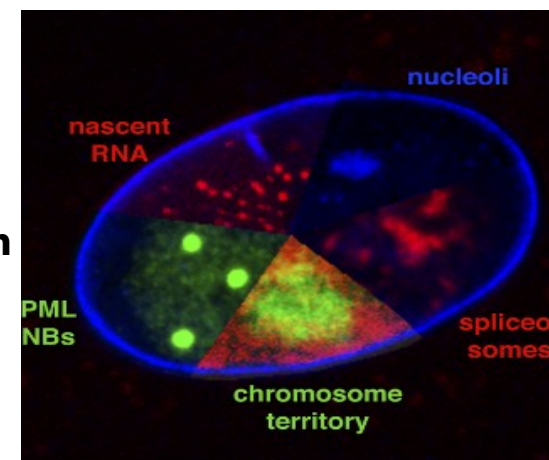


Figure 11-20b
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W.H. Freeman and Company

**Nuclear
organisation**



Brownian motion into equations

Albert Einstein has calculated the diffusion law for a particle :

$$\langle r(t)^2 \rangle = 2nDt$$

$\langle r(t)^2 \rangle$ is the mean squared displacement
 n is the dimensionality
 D is the diffusion coefficient in m^2/s .
 t the time interval between two consecutive steps

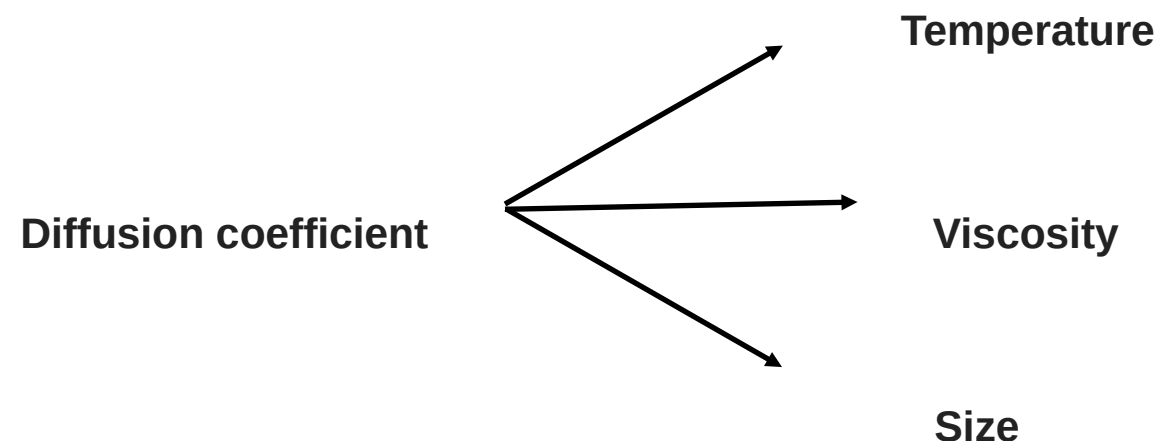
For a truly diffusing particle the MSD scales linearly with time

For isotropic and unrestricted translational diffusion of a spherical particle :

Stoke-Einstein equation :

$$D = \frac{k_B T}{3 \cdot \pi \cdot \eta \cdot d}$$

D is the diffusion coefficient in m^2/s ,
 k_B is the Boltzmann constant,
 T the temperature,
 η the viscosity of the medium and
 d the diameter of the diffusing particle.



Getting to know particles' diffusion coefficients

FRAP/FCS/RICS for diffusion coefficients determination

FRAP

Fluorescence recovery through fluorophores diffusing back within a pre-bleached diffraction-limited volume

FCS

Correlation analysis of the temporal fluorescent intensity of molecules diffusing within a diffraction-limited volume

Limitations :

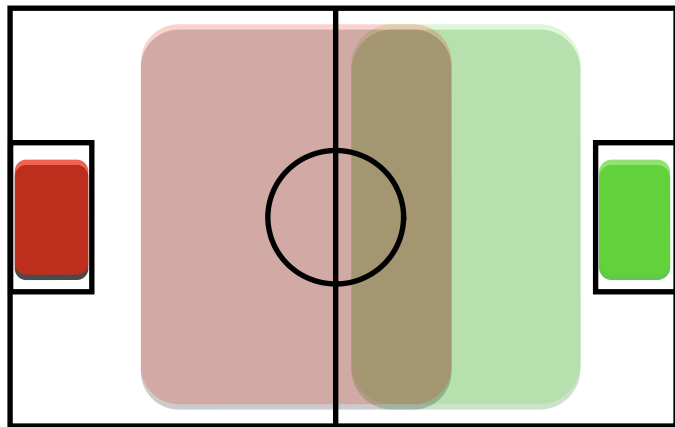
- molecules are not synchronized
 - average properties
- spatial resolution limited to diffraction limit
 - photo-damage

—> go to single molecule conditions

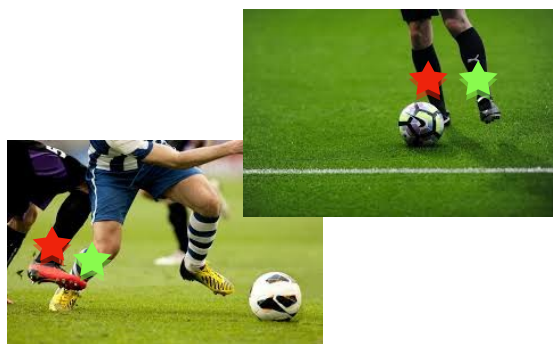
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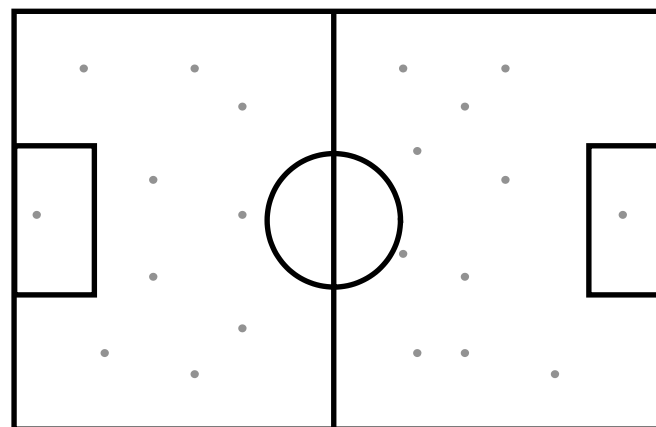
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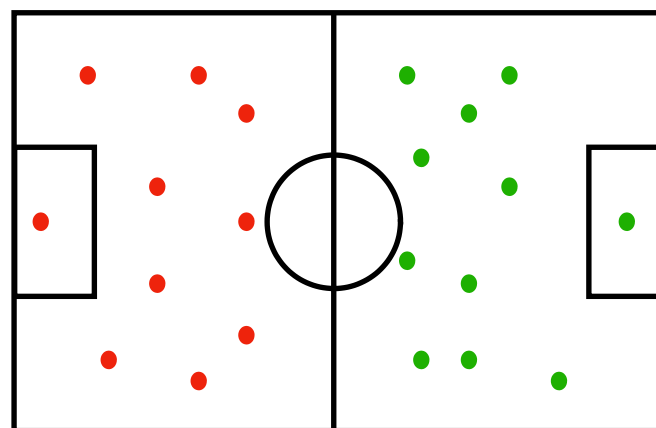
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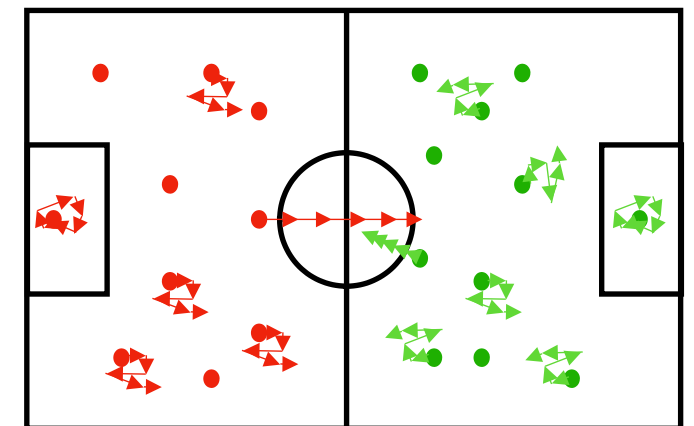
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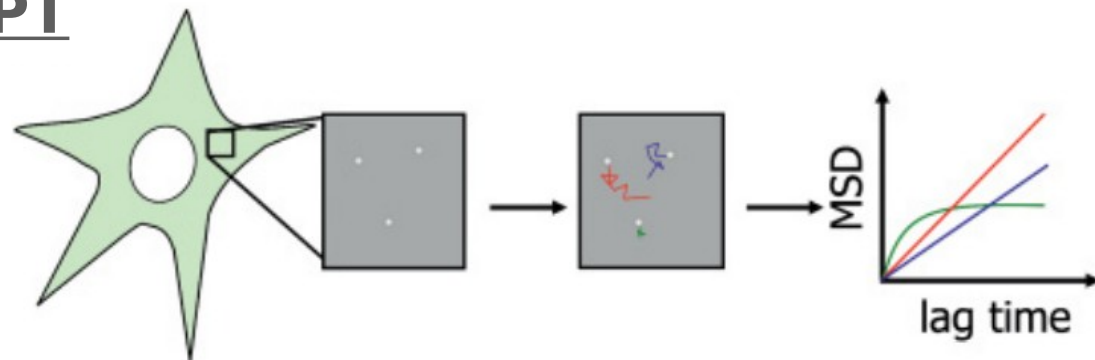


Single Particle Tracking



Single particle tracking

SPT



Direct observation of individual particles as function of time

- no synchronisation
- temporal evolution of each particle (interactions, switches)
 - access populations
 - Rare events

Better spatial resolution

Analysis of single particle tracks can help differentiate immobile vs passive vs active motions

Single particle tracking applications

- Intra- and inter-cellular diffusion of biomolecular structures (e.g. proteins, DNA, lipid vesicles) [1-3]
- Translocation of molecular motors along microtubules [2]
- Organization and transport of lipids, lipid domains, and proteins in the plasma membrane [1]
- Transport of substrates through nuclear pore complexes [7]
- Protein-protein interactions, conformational changes, and ligand binding events using FRET [4]
- Gene expression [2]
- Transcription factor binding [2]
- Virus internalization, signalling, and transport in cells [1, 2, 8]
- Assembly and disassembly of the actin cytoskeleton [9]

• ...

References

1. Manzo, C. and Garcia-Parajo, M.F., *A review of progress in single particle tracking: from methods to biophysical insights. Reports on Progress in Physics*, 2015. 78(12): p. 29.
2. Joo, C., et al., *Advances in single-molecule fluorescence methods for molecular biology. Annual Review of Biochemistry*. 2008, Annual Reviews: Palo Alto. p. 51-76.
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4. Ha, T. and Tinnefeld, P., *Photophysics of Fluorescent Probes for Single-Molecule Biophysics and Super-Resolution Imaging. Annual Review of Physical Chemistry*, 2012. 63: p. 595-617.
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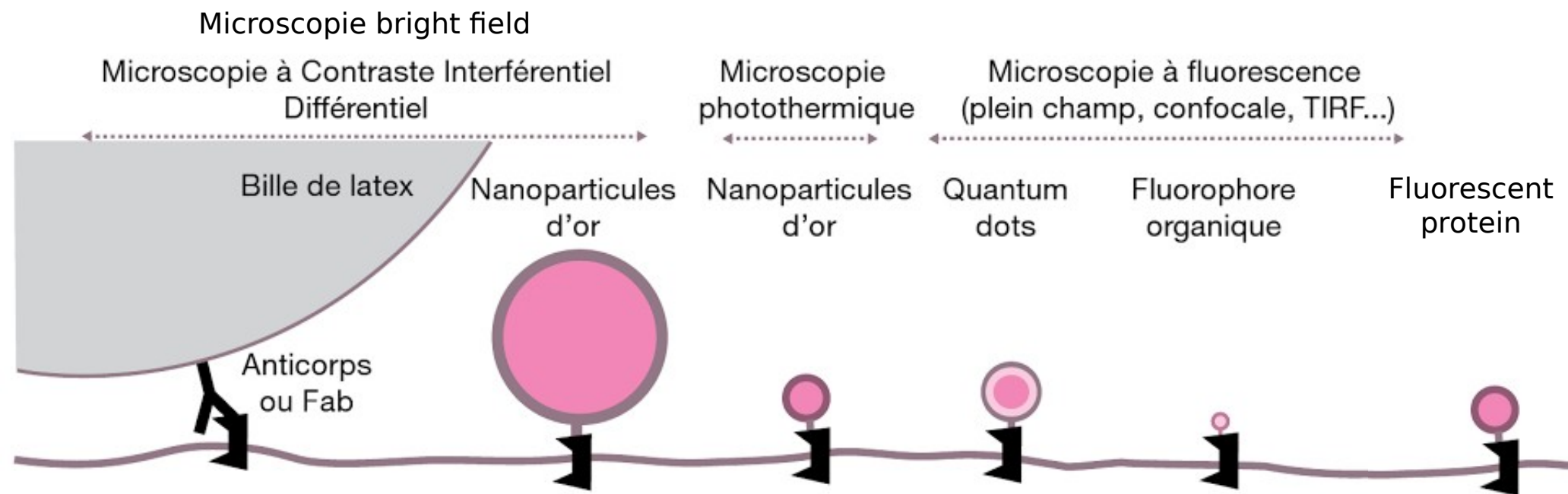
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Imaging approaches

Ideal probe:

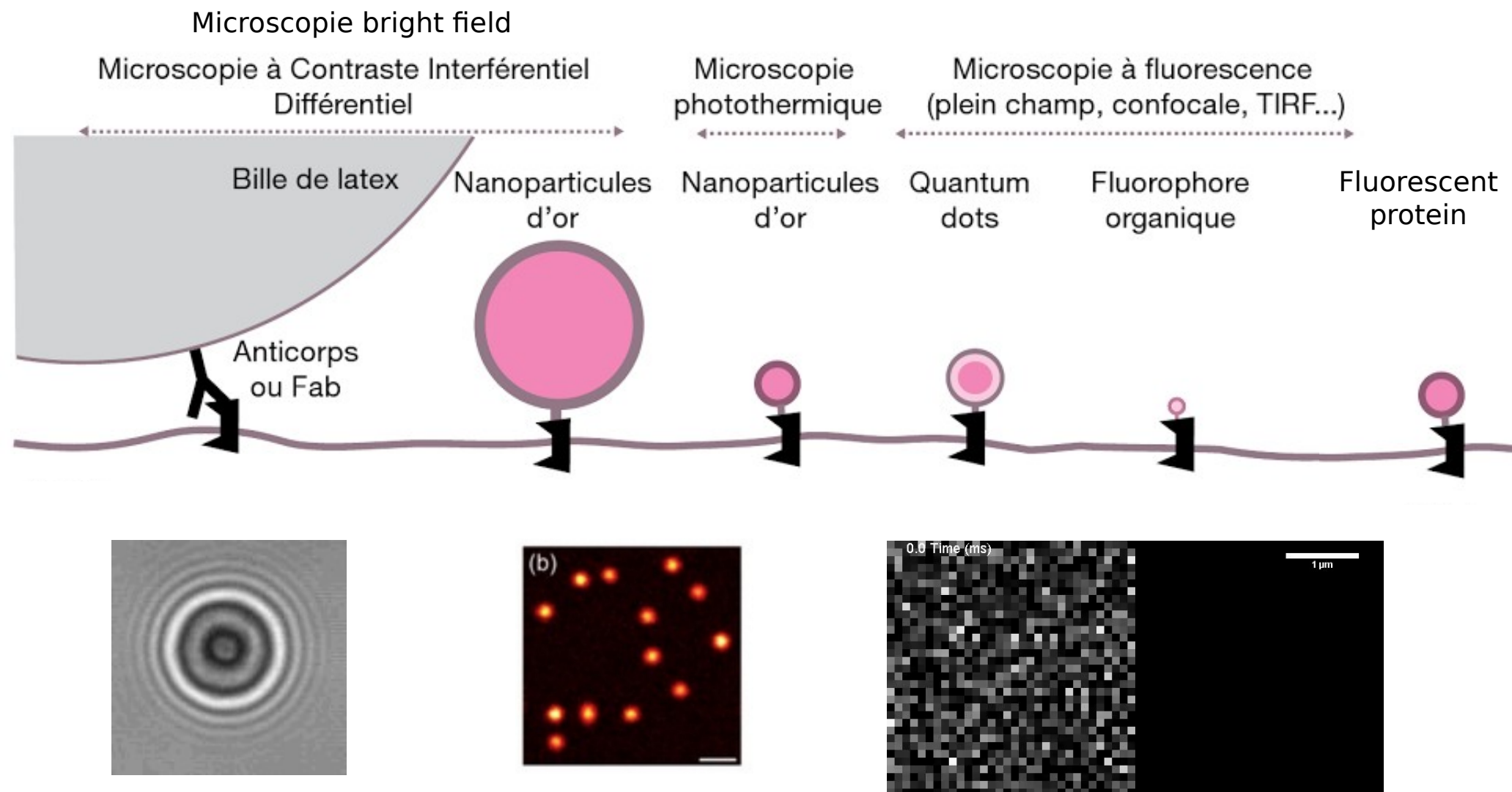
- Small compared to the molecule of interest (does not interfere with its motility)
 - Specific (doesn't stick everywhere) and doesn't aggregate
 - Efficient (labelling ratio 1:1)
 - Photo-stable (bright, no blinking and no photobleaching)



Imaging approaches

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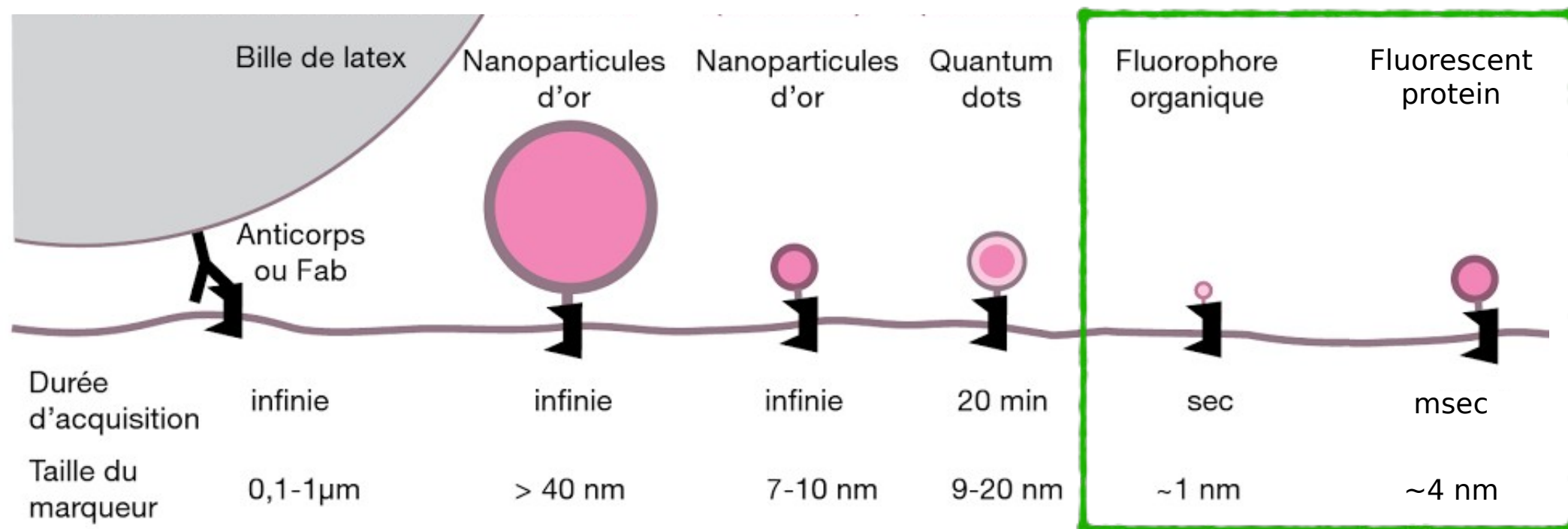
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Probes

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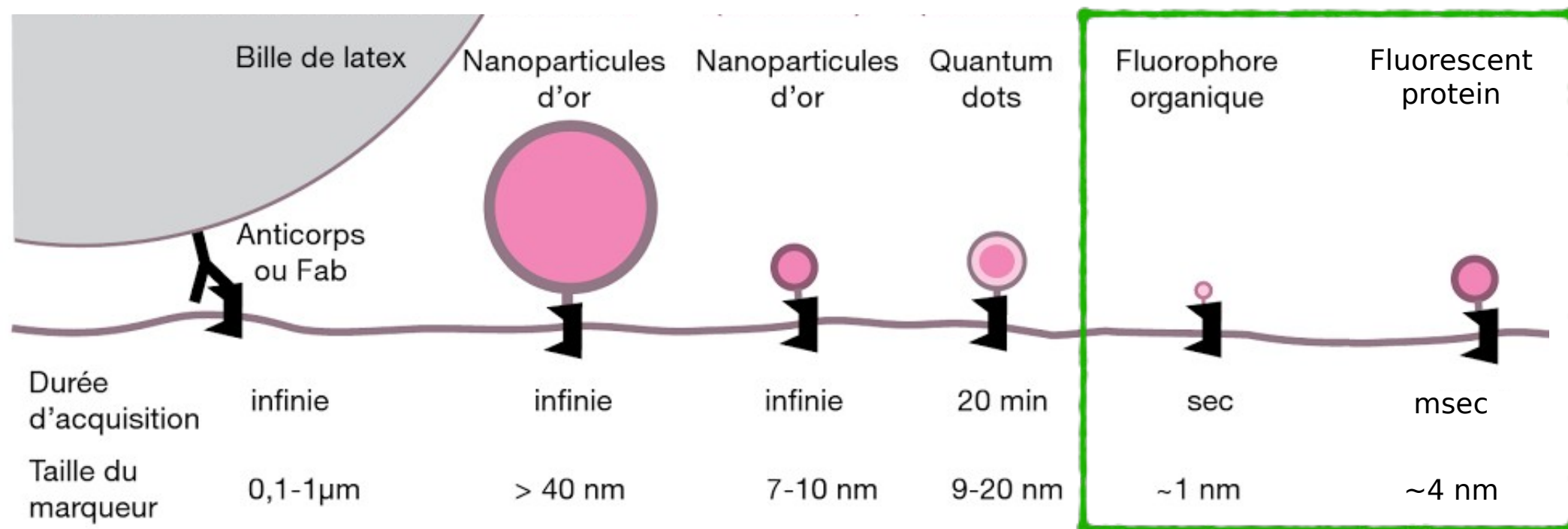


- **difficulty of delivering fluorescent probes past the membrane barrier in live cells**
 - **molecules must be efficiently isolated to be localized precisely**
 - ...

Probes

Labelling density in single-molecule microscopy also plays into the issue of **cellular toxicity** and/or **non-physiological behavior**.

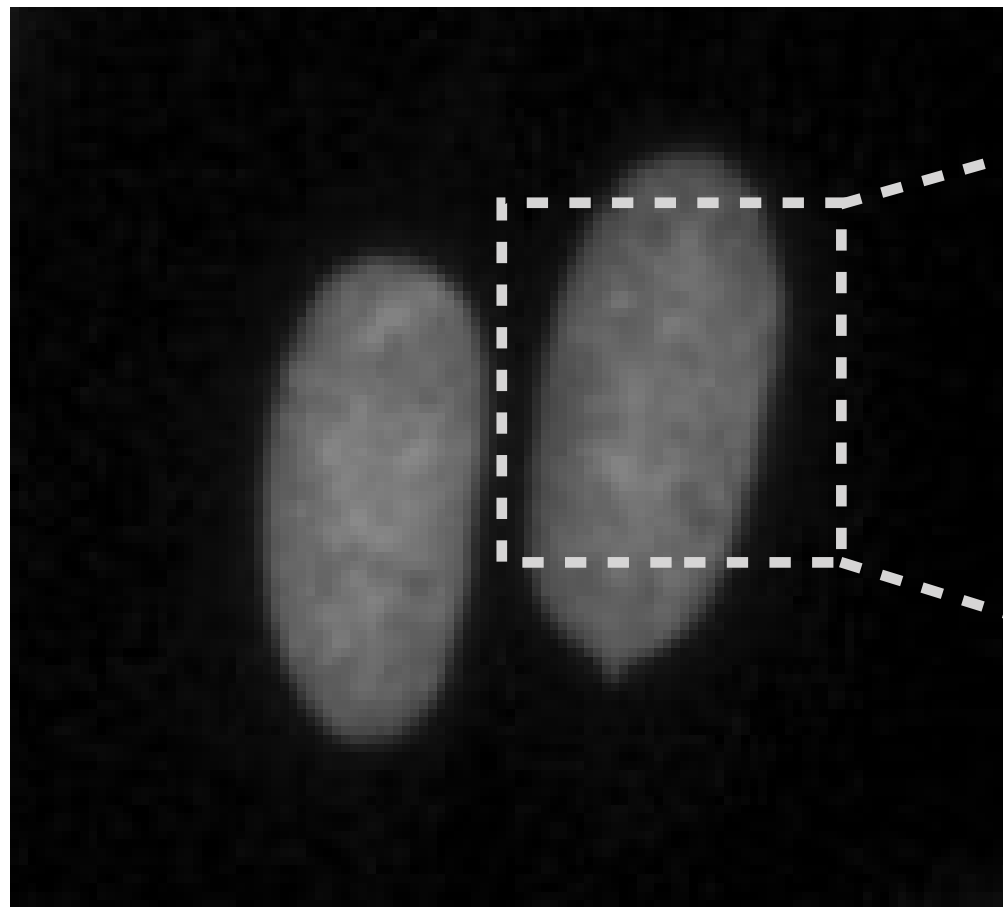
- stress of introducing exogenous labels
 - overexpression of FPs
 - protein aggregation
 - impaired protein function
 - ...



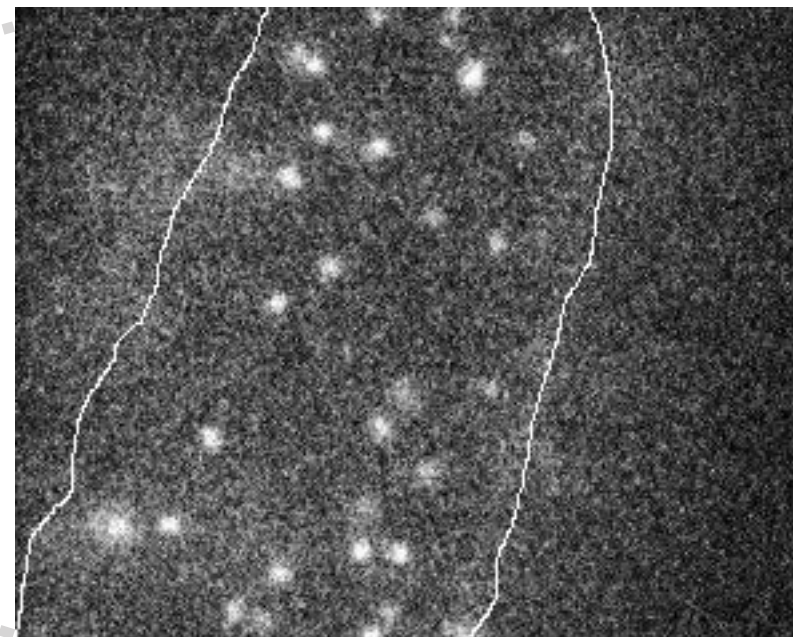
- **difficulty of delivering fluorescent probes past the membrane barrier in live cells**
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 - ...

Getting to single molecule conditions

What you'll get



What you want



⇒ **Need to get to single molecule conditions**

Getting to single molecule conditions

For SPT experiments, the density of standard FP- and dye-based labels is often initially too high to isolate single molecules.

⇒ Partial labeling

Limited statistics per cell.

⇒ Prebleaching to reduce the concentration to where single-molecule imaging is possible

*Limited statistics per cell.
Contribute significant initial stress on the cell*

⇒ Use photo-activable probes (spt-PALM)

Prolonged exposure to weak (typically $<1 \text{ W/cm}^2$) near UV light for activation and intense ($>1 \text{ kW/cm}^2$) visible light for single molecule excitation can also produce significant damage

⇒ **Either standard FP- and dye-based labels or photo-activable labels**

FP- and dye-based labels

	Fluorescent proteins	Organic dye
Standard	⇒ Object of interest is isolated ⇒ Partial labeling ⇒ Prebleaching	
Light-controllable	⇒ Light to isolate single molecules	

Fluorescent protein tagging

photostable FPs

Genetically Encoded fluorescent proteins

Common multicolour

filter sets :

CFP/ YFP/ and RFP

DAPI / Fluorescein

Rhodamine / Cy5



Name	λ_{ex} (nm)	λ_{em} (nm)	E (mM ⁻¹ cm ⁻¹)	QY	$E \times QY$	Type	pK _a	Bleach time (s)	Reference
Intrinsically fluorescent proteins									
mTagBFP2	399	454	51	0.64	32.4	m	2.7	53	Subach et al. (2011)
mTurquoise2	434	474	30	0.93	27.9	m	3.1	90	Goedhart et al. (2012)
mCerulean3	433	475	40	0.80	32	m	4.7		Markwardt et al. (2011)
EGFP	488	507	56	0.60	33.6	m	6.0	174	Yang et al. (1996)
mWasabi	493	509	70	0.80	56	m	6.0	93	Ai et al. (2008)
Superfolder GFP	485	510	83	0.65	54.1	m			Pédelacq et al. (2006)
mNeonGreen	506	517	116	0.80	92.8	m	5.7	158	Shaner et al. (2013)
mClover3	506	518	109	0.78	85	m	6.5	80	Bajar et al. (2016b)
Venus	515	528	92	0.57	52.5	m	6.0	15	Nagai et al. (2002)
Citrine	516	529	77	0.76	58.5	m	5.7	49	Griesbeck et al. (2001)
mKOκ	551	563	105	0.61	64	m	4.2		Tsutsui et al. (2008)
tdTomato	554	581	138	0.69	95.2	td	4.7	98	Shaner et al. (2004)
TagRFP-T	555	584	81	0.41	33.2	m	4.6	337	Shaner et al. (2008)
mRuby3	558	592	128	0.45	57.6	m	4.8	349	Bajar et al. (2016b)
mScarlet	569	594	100	0.70	70.0	m	5.3	277	Bindels et al. (2017)
FusionRed	580	608	95	0.19	18.1	m	4.6	150	Shemiakina et al. (2012)
mCherry	587	610	72	0.22	15.8	m	4.5	96	Shaner et al. (2004)
mStable	597	633	45	0.17	7.6	m		1002	Ren et al. (2016)
mKate2	588	633	63	0.40	25	m	5.4	84	Shcherbo et al. (2009)
mMaroon1	609	657	80	0.11	8.8	m	6.2	178	Bajar et al. (2016a)
mCardinal	604	659	87	0.19	16.5	m	5.3	730	Chu et al. (2014)
T-Sapphire	399	511	44	0.60	26.4	m	4.9	25	Zapata-Hommer and Griesbeck (2003)
mCyRFP1	528	594	27	0.65	18	m	5.6	45	Laviv et al. (2016)
LSSmOrange	437	572	52	0.45	23.4	m	5.7		Shcherbakova et al. (2012)
mBeRFP	446	611	65	0.27	17.6	m	5.6		Yang et al. (2013)

old

- Spectral choices, multicolour
- Expressed endogenously with protein of interest
- Simplicity
- Specificity

- Ok if object of interest is isolated, otherwise:
- Limited statistics per cell.
- Contribute significant initial stress on the cell (if bleaching to get isolated molecules)

Fluorescent protein tagging

Genetically Encoded light-controllable fluorescent proteins

PAFP	Preactivation/postactivation emission wavelength, nm*	Photon no.	ClpP clustering [†]	No. of localizations per cell [‡]	Maturation time, min [§]	Photostability...
Dendra2	507/573	686	—	1,810	38	
mEos2	519/584	745	+	1,290	340	
mEos3.2	516/580	809	—	1,950	330	
tdEos	516/581	774	—	1,800	330	
mKikGR	515/591	599	+	3,800	31	
PAmCherry	—/595	706	+	4,200	61	
PAtagRFP	—/595	906	—	760	200	
mMaple	505/583	798	+	24,000	48	
mMaple2	506/582	783	+	21,000	62	
mMaple3	506/583	675	—	12,300	49	
PAGFP	—/517	313	—		<10	
PSCFP2	468/511	223	+			
Dronpa	—/517	262	—		25	
mGeosM	—/514	248	+		<10	

better localisation
precision/longer
tracks

Protein
aggregation/o
ligomerisatio
n

signaling
efficiency/protein
folding efficiency

Expressed endogenously with protein of interest

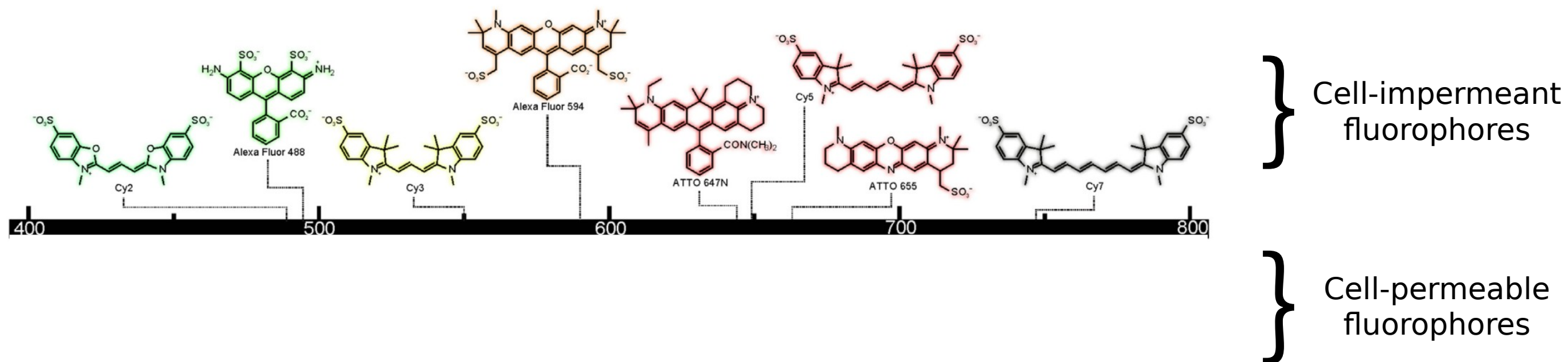
- Simplicity
- Specificity
- ...

- **Limited choice in PA-FP spectral properties**
- **Very limited tracking # of frames (~<20 frames)**
 - Low folding efficiencies
 - Stress induced photo activation
 - ...

Organic dye labelling

Introducing labeled proteins or oligonucleotides into live cells is difficult, typically requiring microinjection

Development of highly photostable cell-permeable organic dyes → Janelia Farm dyes



Now available as photo-activable!

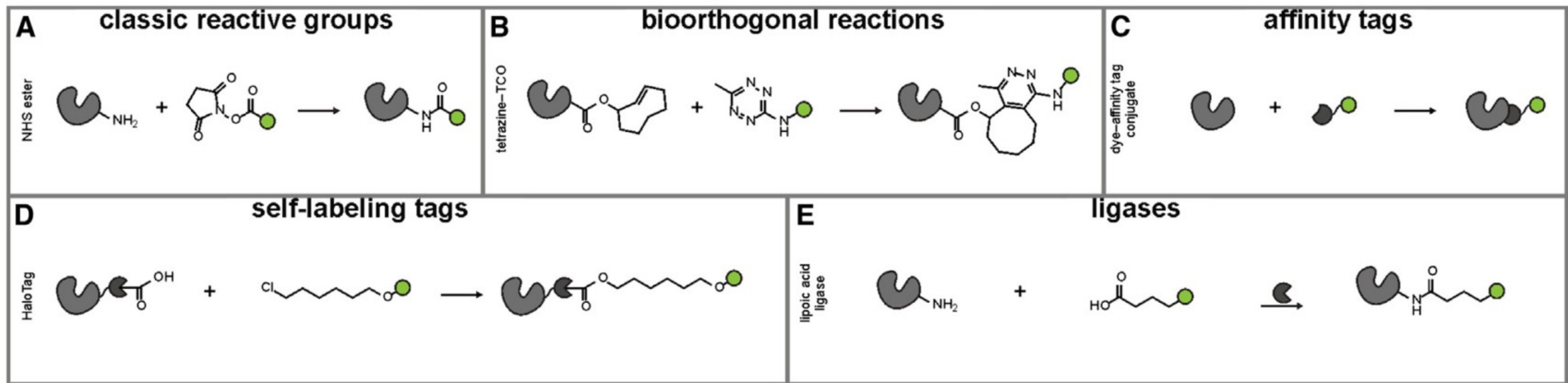
Grimm et al, Nat Meth, 2016

- spectral choices, multicolour
- brightness,
- photostability,

- sophisticated approaches to increase labelling specificity and to deliver them to biological cells

Organic dye labelling

—> Development of genetically encoded, protein-specific attachment sites in live cells (e.g., Snap-tag and HaloTag)



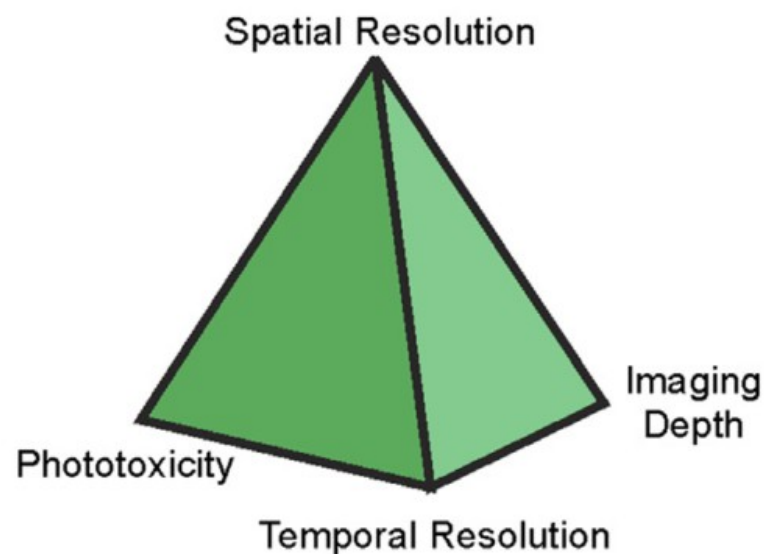
Genetic encoding—one of the main advantages of fluorescent proteins—with the favora

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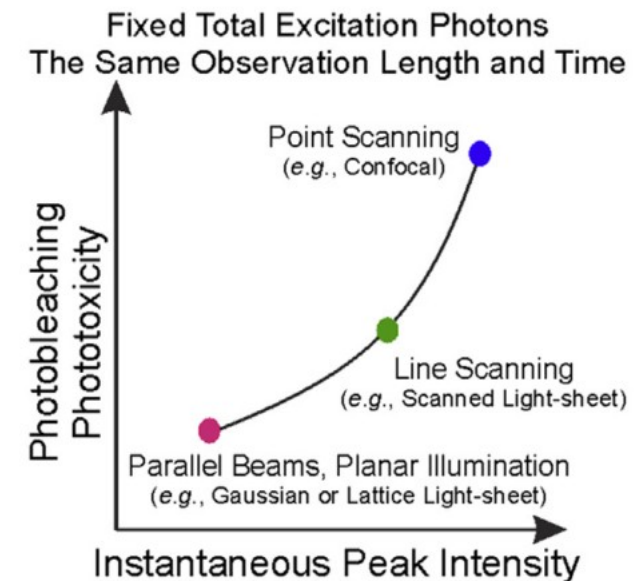
General considerations in live-cell imaging

Imaging trade-offs

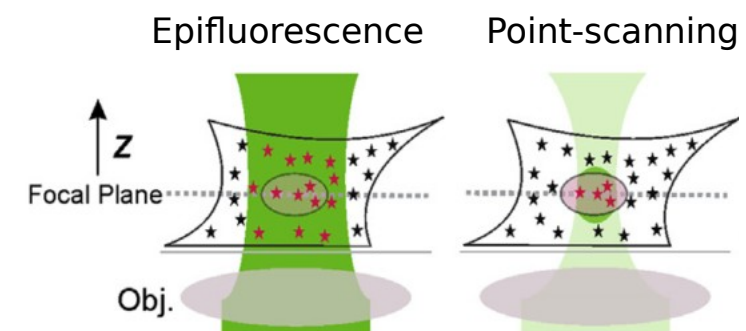


Phototoxicity : Excitation of molecules such as chromophores inside the cell can generate reactive oxygen species (ROSs) that can damage a variety of cellular constituents (e.g., protein, DNA, lipid) and either alter the biological process of interest or elicit cell-cycle arrest or cell death.

Point vs uniform illumination



- 1- uniform illumination induces less photodamage compared to point- or line-scanning methods
- 2- due to the relatively slow point-scanning acquisition, confocal microscopy is limited in its ability to detect fast molecular dynamics

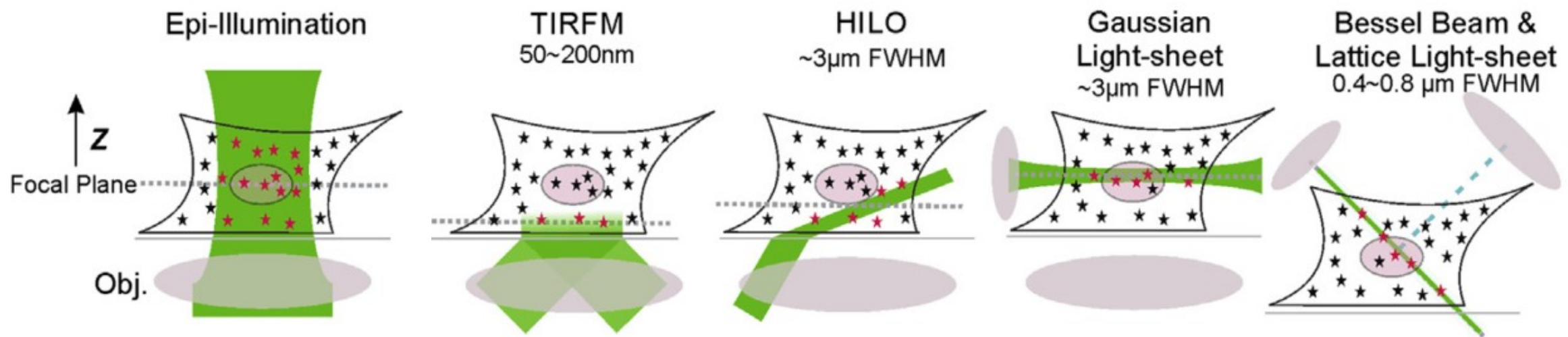


Illumination

- ⇒ Maximize signal to noise ratio
- ⇒ reduce photo-toxicity

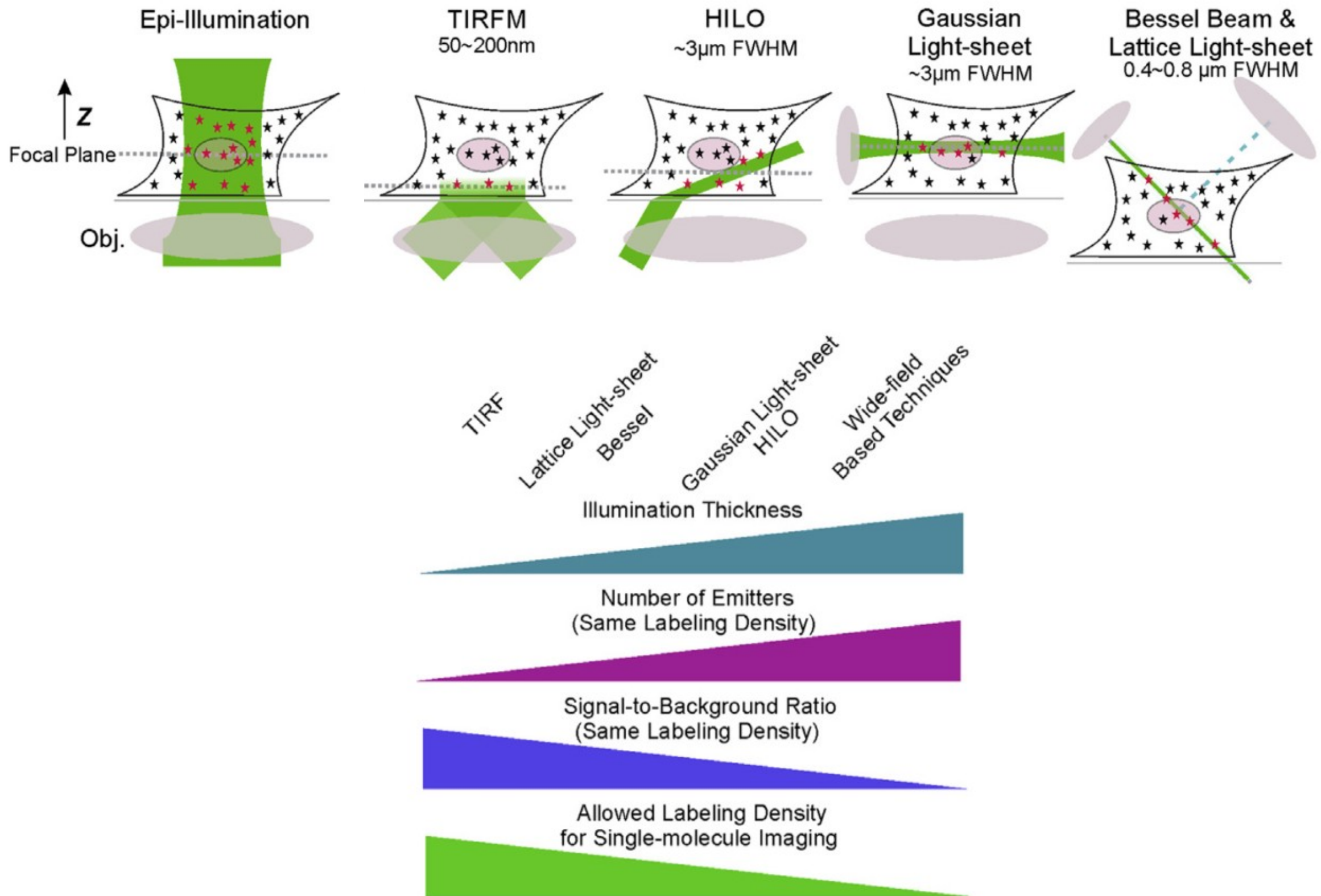
Out-of-focus molecules :

- 1- contribute to background, lowering the SBR of single-molecule detection and decreasing the localization precision
- 2- can bleach prematurely before they can be localized



Beware of limited depth of field and depth of excitation
Diffusing molecules may get away from detection/excitation area

Illumination



Acquisitions

- ⇒ **minimise photobleaching** (adjust illumination intensity, add photo stabilisers if applicable, choose the right fluorophore, ...)
- ⇒ **minimise photo-toxicity** (adjust illumination intensity, choose the appropriate illumination approach, ...)

Molecules of interest is moving fast

- ⇒ Use faster camera (sCMOS)
- ⇒ Use stroboscopic illumination/acquisitions
 - ⇒ Decrease camera integration time
 - ⇒ Use very short & intense excitation bursts

Motion blurring

Motion blurring

$$D_{\max(2D)} = \frac{\Delta^2}{4\delta t}$$

$D < D_{\max}$
Fair SM Detection



$D \gg D_{\max}$
Motion Blur



Molecules of interest is moving slowly

- ⇒ Use slower acquisitions to minimise photo-toxicity
 - ⇒ longer / less frequent camera integration time
 - ⇒ lower excitation power

Spatial displacement within camera's acquisition time should be smaller than the localization

Acquisition is long / tracking over long timescales

- ⇒ Sample drift correction (fiducial markers)
- ⇒ Autofocus
- ⇒ Stroboscopic illumination
- ⇒ Space in time your images and laser excitations

Multi colors acquisitions

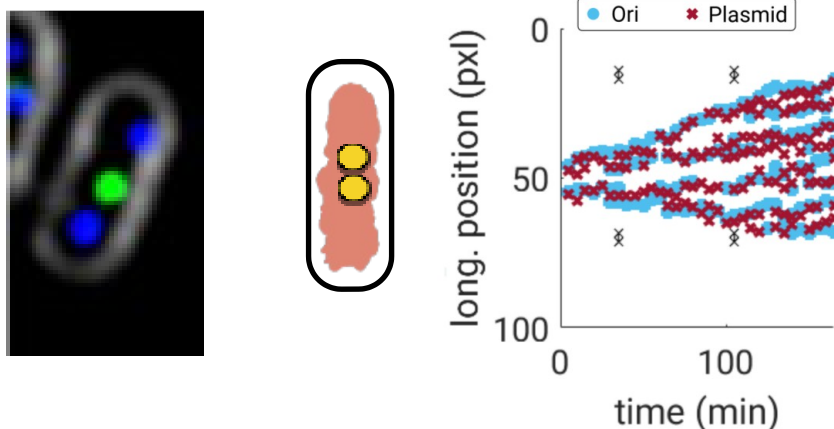
- **powerful tool to track and quantify interactions between the same species or between multiple components.**

Neurotransmitters in live neurons

- **Spatial and temporal correlation analysis**
 - **Co-localization**
 - **Interactions**
 - ...

Cognet lab @ LP2N

Chromosome organisation and segregation



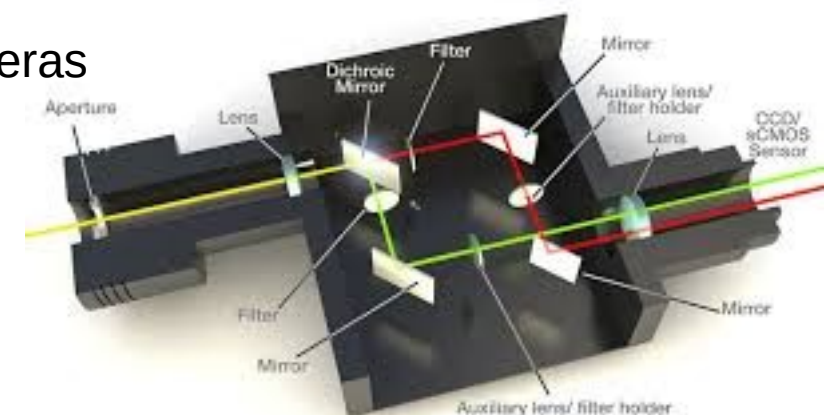
Nollmann lab @ CBS

Fiducial markers to align images

Alternating excitation lasers to reduce cross-talk

between fluorescent signals

Dual-view / 2 cameras



Outline

- 1.** Introduction on brownian motion
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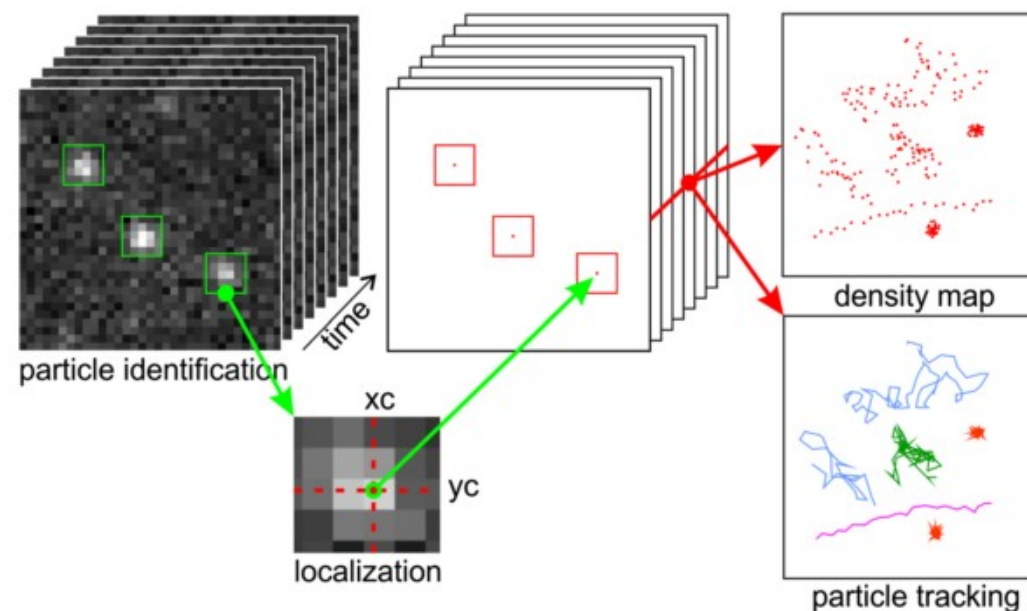
Outline - Part 1 : Analysis

Particle identification and
Localization

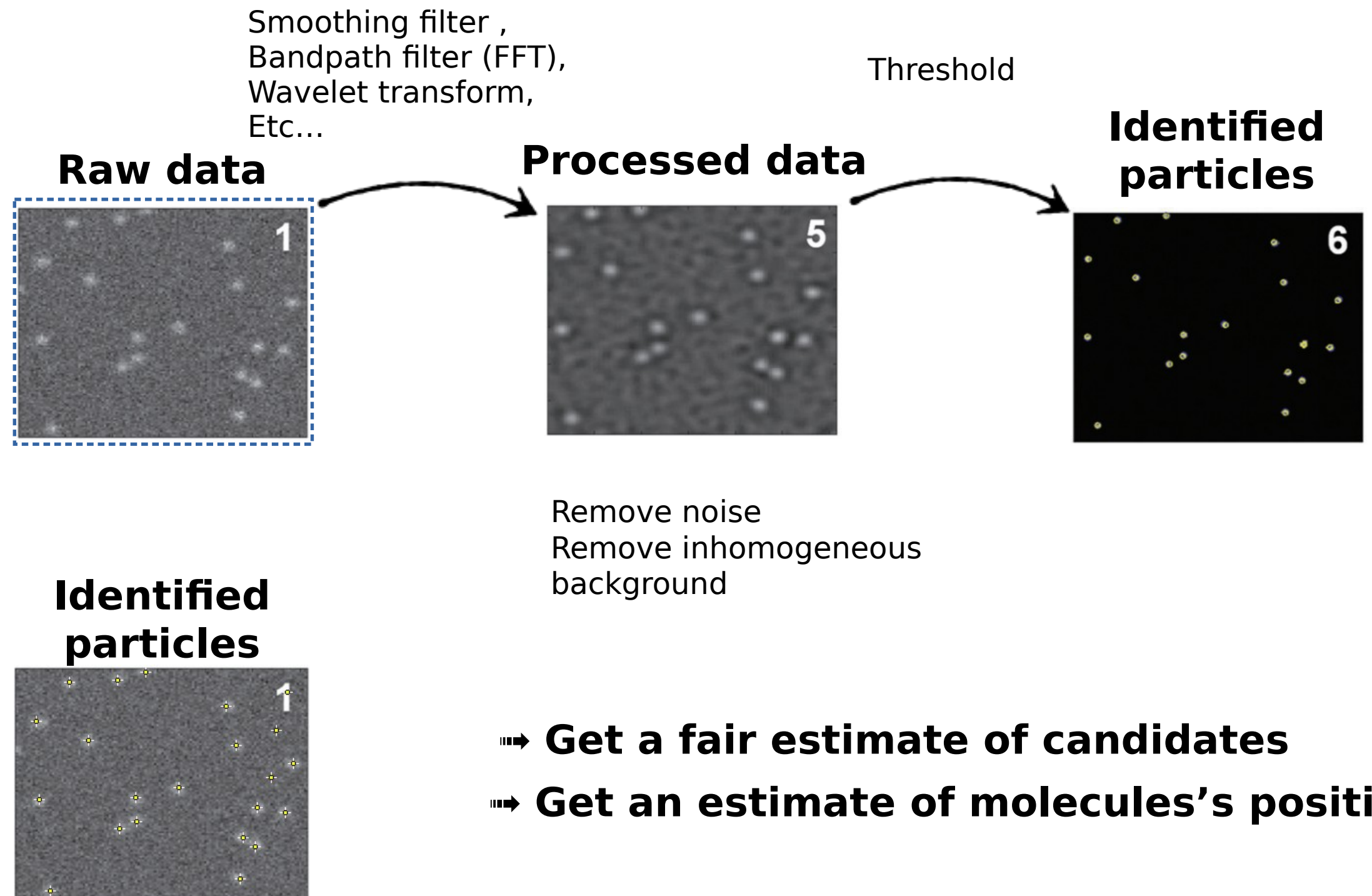
Reconnecting data points

Localization error

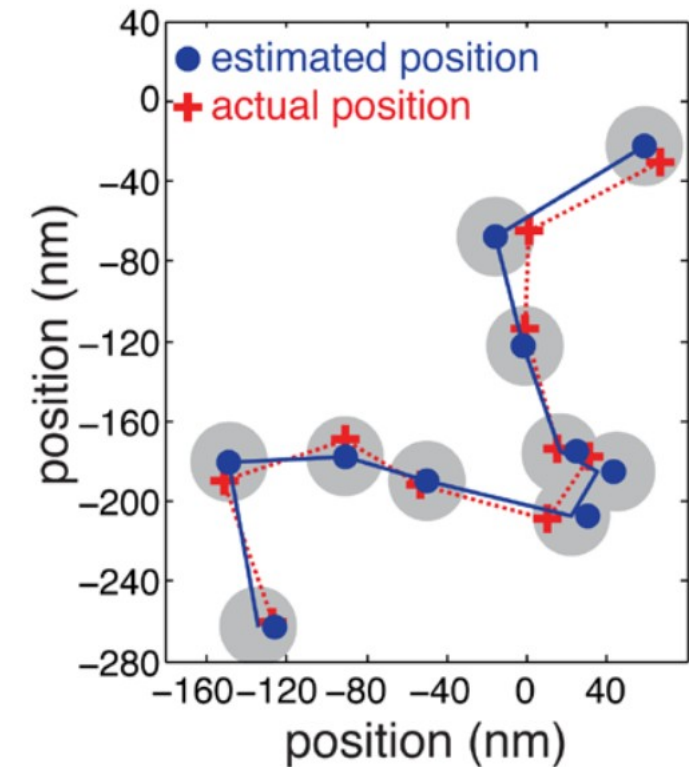
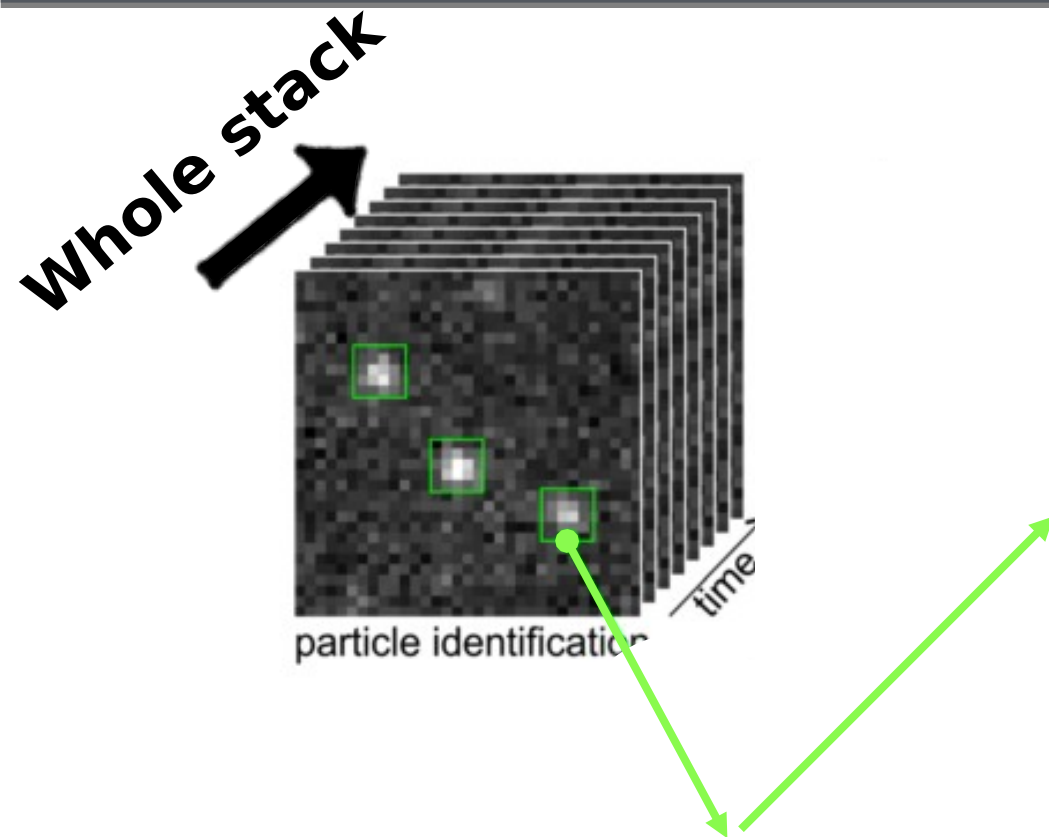
Mean Squared Displacement (MSD)



Particle identification

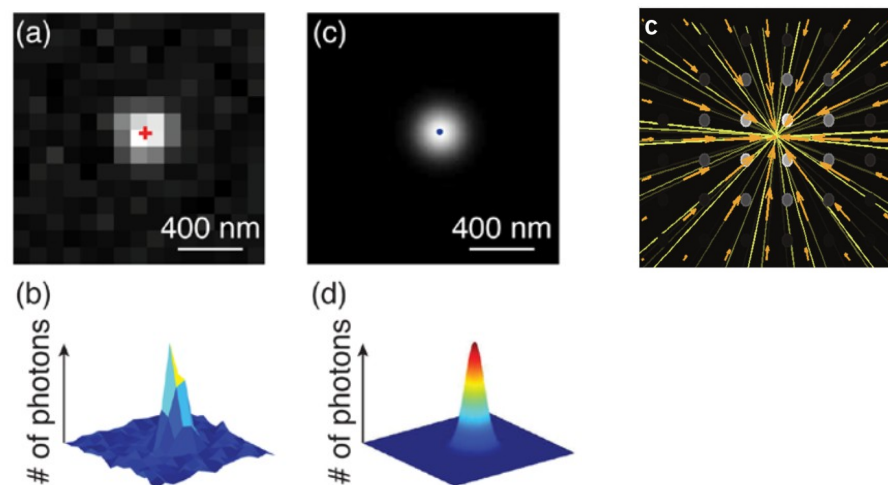


Particle localisation



Particle precise localization data

Control / PSF fitting / Radial symmetry / etc...

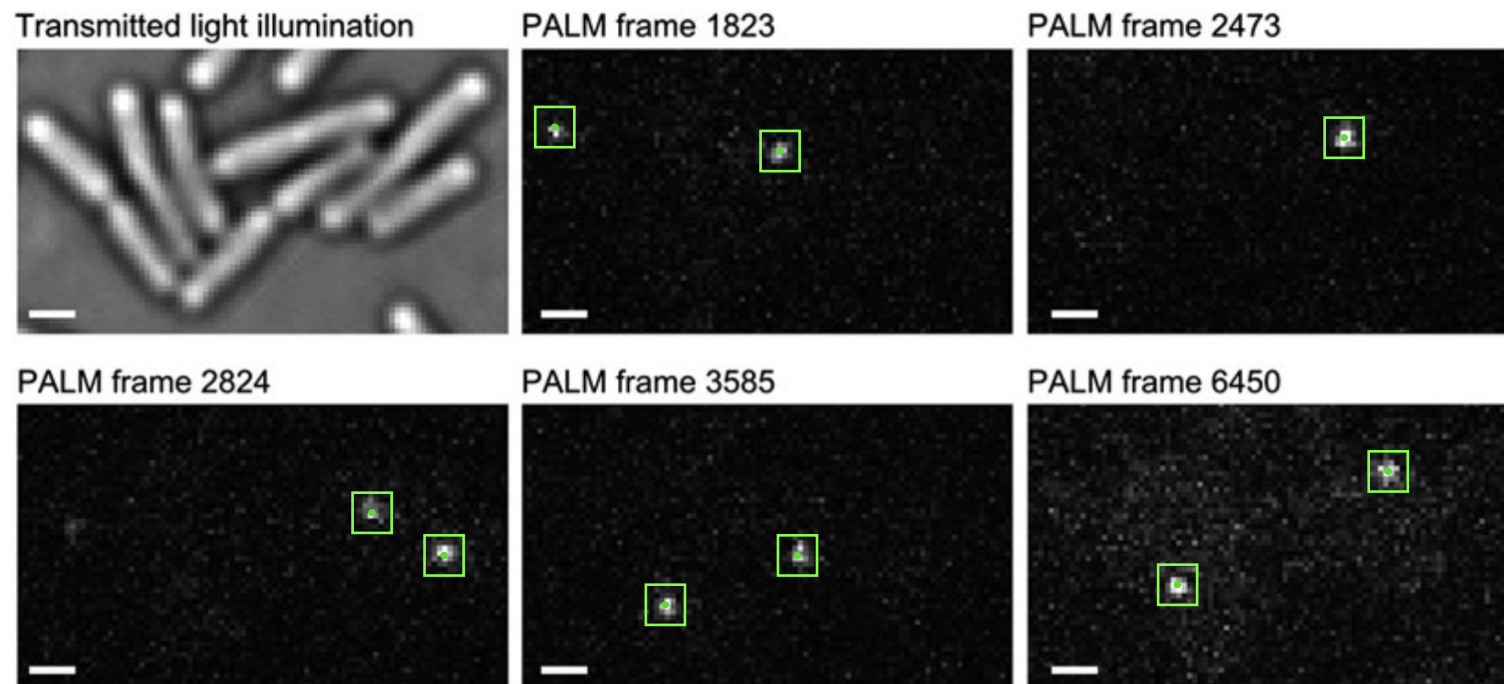


Localization errors

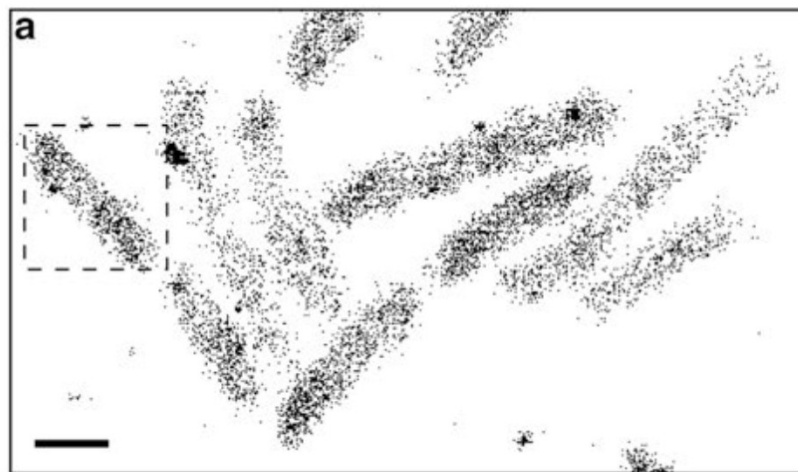
- Estimation errors (PSF model, detector noise, collected photons, background, etc...)
- Particle motion during the acquisition time produces a change in the shape of the PSF and an inherent decrease in localization precision

Particle localisation - case study - part 1/3

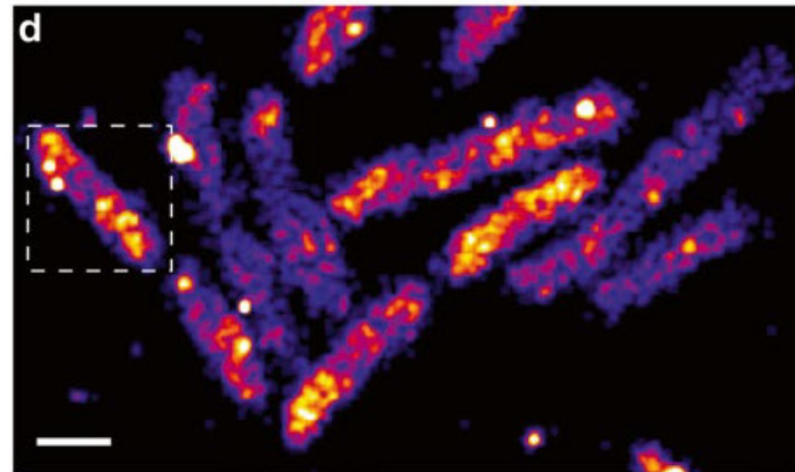
⇒ Single particle tracking PALM (spt-PALM) of XX proteins in E.coli



Localization map



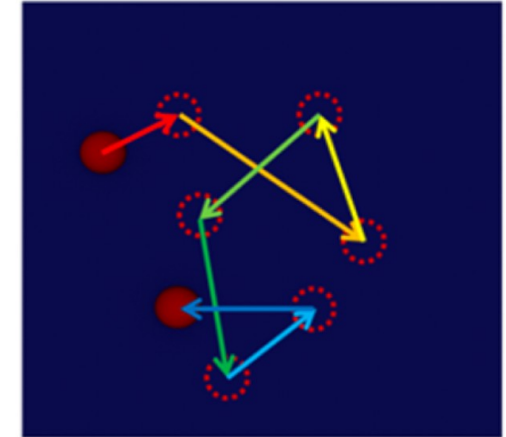
Localization density map



Reconnecting the dots

At ultra-low particle densities : simply join the nearest dots from frame to frame.

At higher densities : ambiguities can arise from path-crossing and particle appearance/disappearance (fluorophore photophysics, false-positive, missed localization, etc...)



Consequences :

⇒ **short, fragmented trajectories**

⇒ **wrongly joining localizations belonging to different particles**

Solutions :

⇒ criteria for stopping a trajectory reconstruction when no localizations are found within a given distance and/or time window

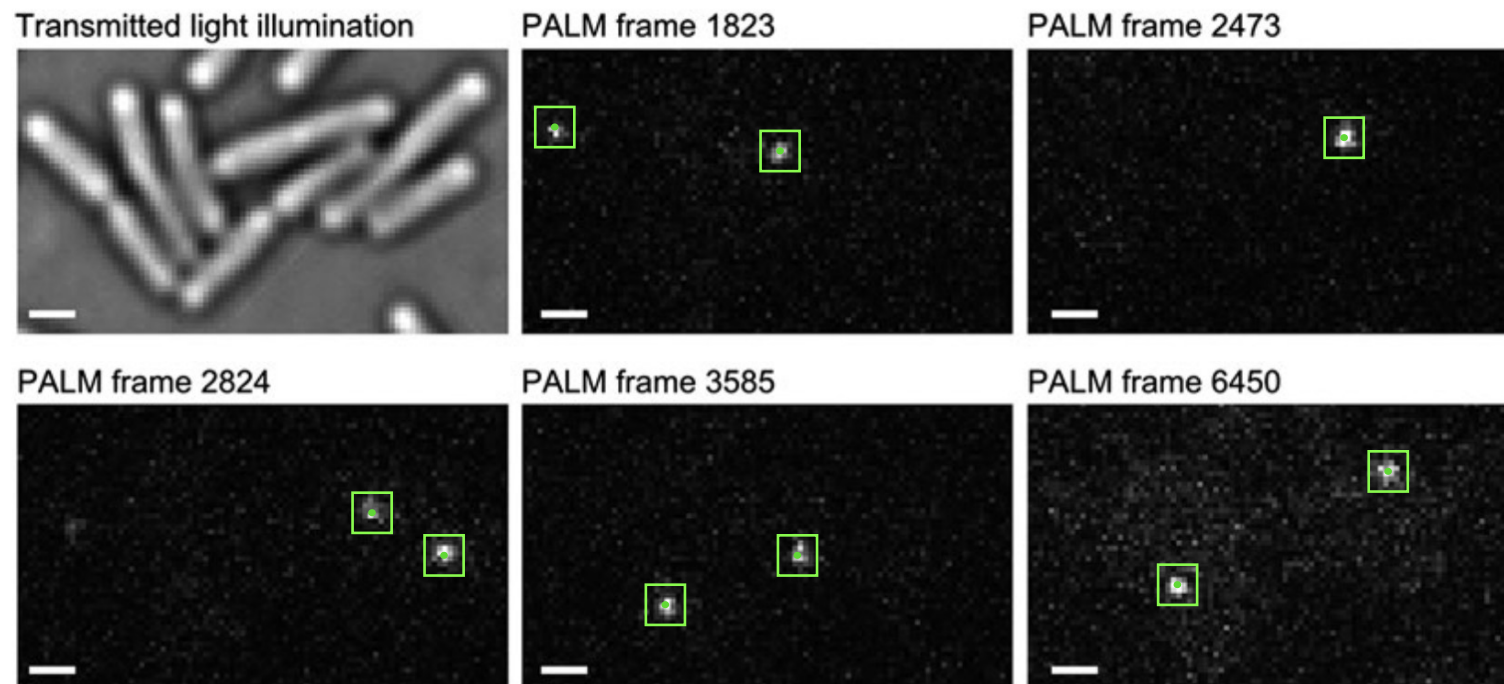
⇒ fluorescence intensity, PSF shape as well as models of the type of particle motion and emitter photophysics can also be included in the linking routine to improve reconnection performances

⇒ prior knowledge about the particle's type of motion is a key factor for tracking reliability

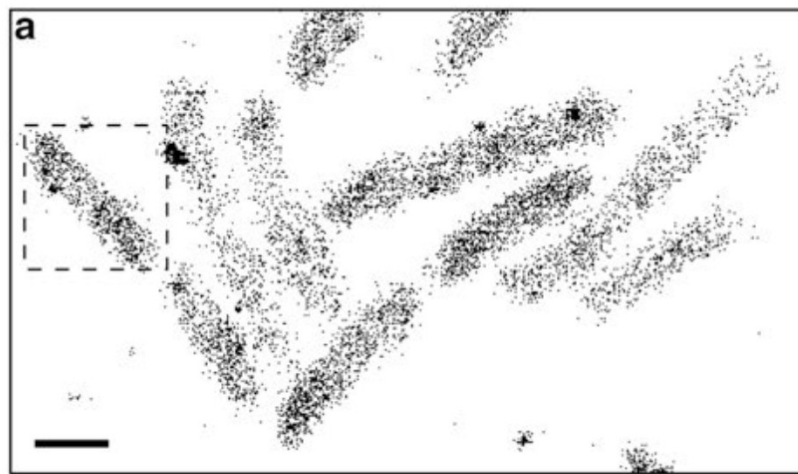
⇒ generate synthetic training-data to tune the algorithm parameters for each experimental condition

Particle localisation - case study - part 1/3

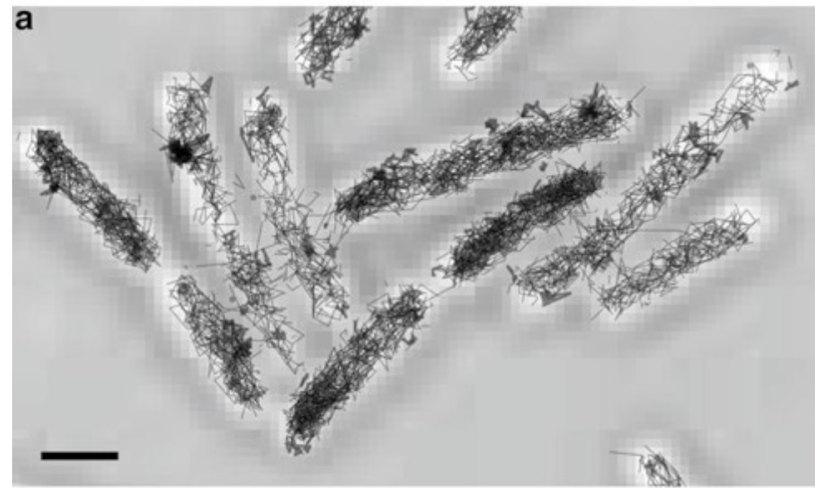
⇒ Single particle tracking PALM (spt-PALM) of XX proteins in E.coli



Localization map



Localization density map



Mean Squared Displacement (MSD)

Particle movements can be measured by the mean square displacement (MSD) and the lag time Δt and is characterized by the diffusion coefficient D which is a measure of the speed of diffusion:

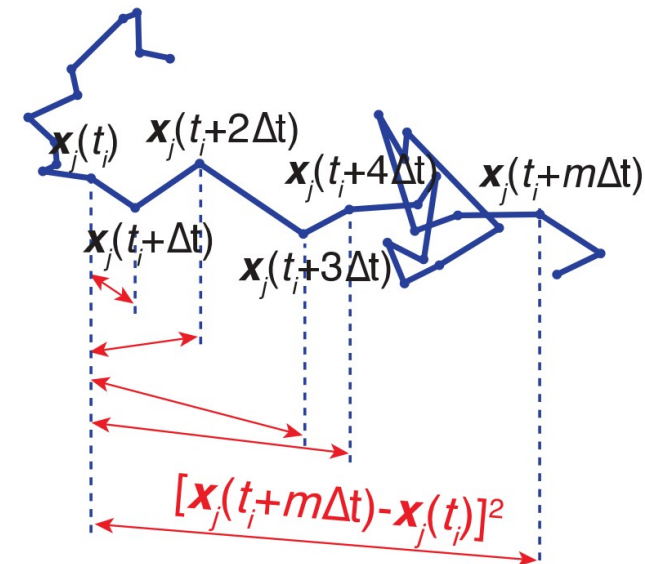
$$\text{MSD}(\Delta t) = \langle r(\Delta t)^2 \rangle = 2.n.D.\Delta t$$

$\langle r(t)^2 \rangle$ is the mean squared displacement

n is the dimensionality

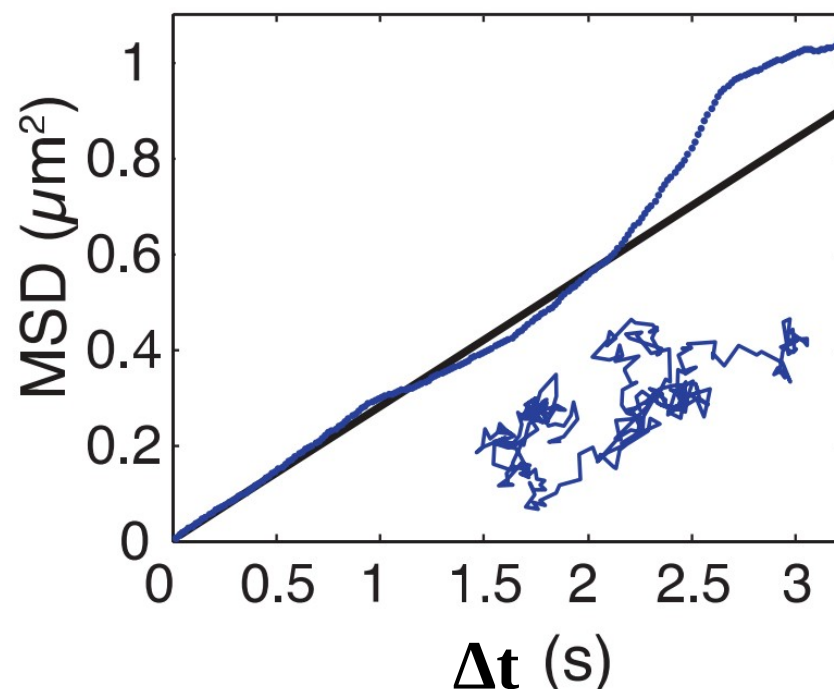
D is the diffusion coefficient in m^2/s .

Δt the time interval between two consecutive steps



Plotting the MSD :

Each data point in the MSD plot is the result of an average over different times Δt :



Extracting the diffusion coefficient:

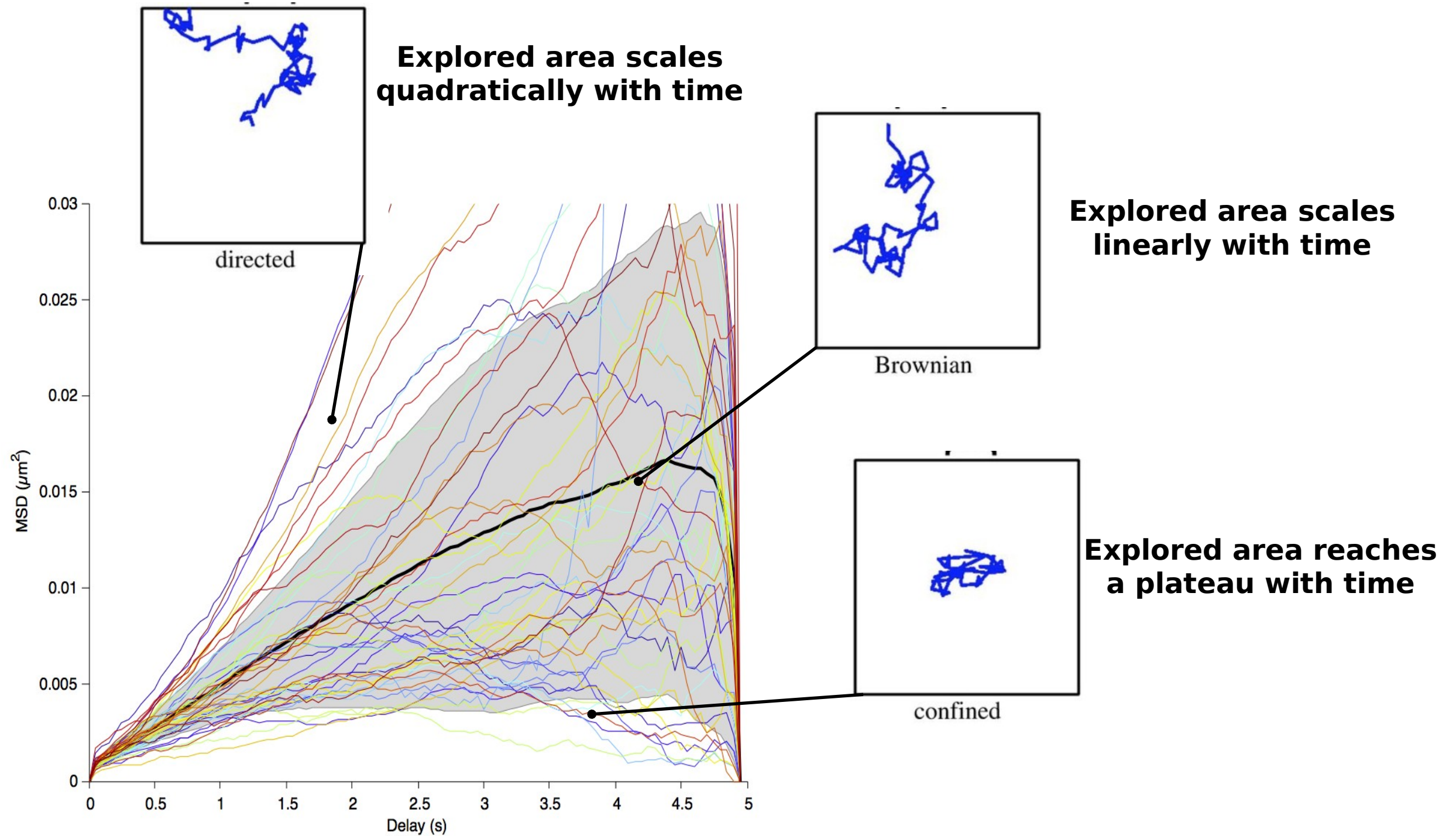
The diffusion coefficient D can be obtained :

- by fitting of the first four points of the MSD using $r^2 = 2nDt$ linear equation.
- by computing the instantaneous diffusion coefficient, which is calculated for each time point of the trajectory by linear fitting of the 4 following points of the MSD.

⇒ If the particle is freely diffusing!

may not be appropriate when a molecule changes its motion.

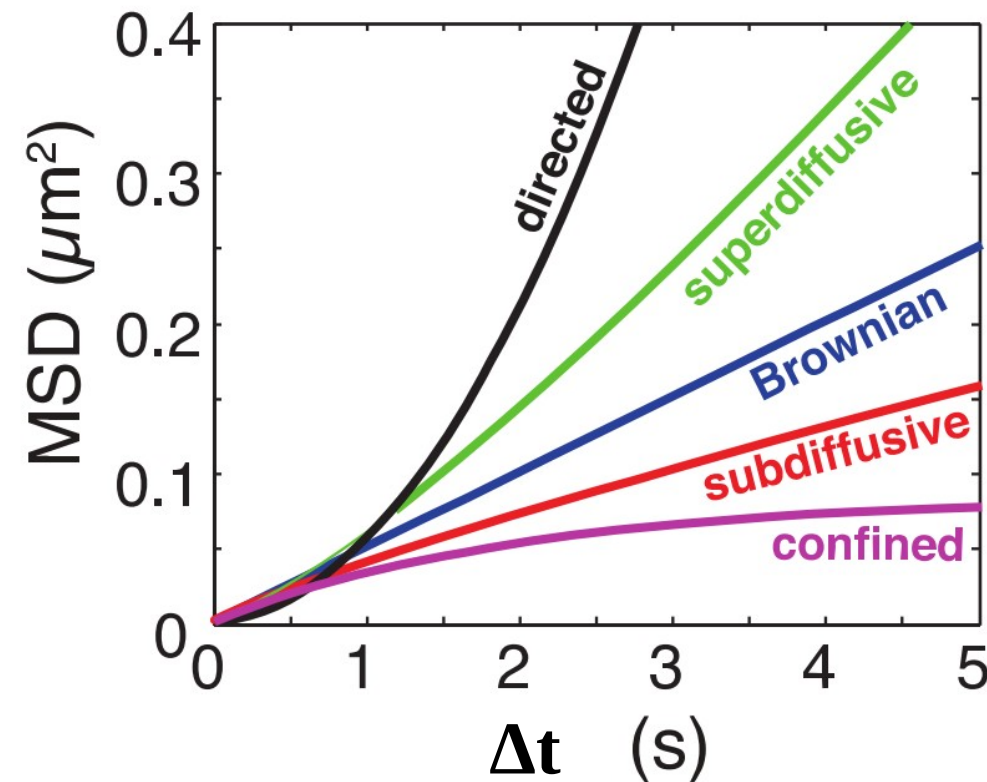
Plotting experimental MSDs



⇒ Most particles AREN'T freely diffusing!

MSD analysis

→ Extract single particle behaviors based on MSD curves



Directed diffusion : ballistic movement or active transport dominating for larger values of t

Anomalous (super or sub-) diffusion can occur due to interactions with obstacles that alter diffusion

Confined motion shows, after an equilibration time, saturation to a characteristic plateau, whose value is proportional to the confinement area.

In 2 dimensions:

Directed movements

$$\langle r^2 \rangle = v^2 t^2 + 4Dt$$

Brownian movements

$$\langle r^2 \rangle = 4Dt$$

Anomalous diffusion

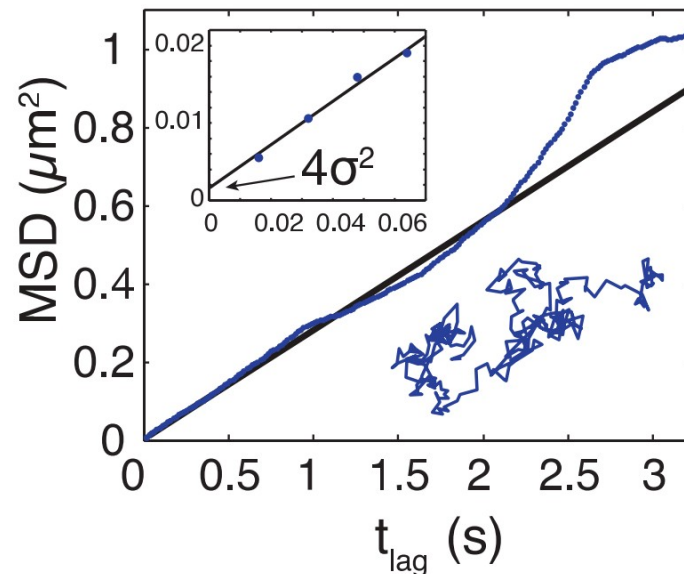
$$\langle r^2 \rangle = 4Dt^\alpha, 0.1 \leq \alpha \leq 0.9 \text{ or } \alpha > 1$$

Confined movements

$$\langle r^2 \rangle = \frac{4R^2}{3} + \frac{128R^2}{\pi^4} \sum_{n=1, \text{odd}}^{\infty} \frac{1}{n^4} \exp\left(-\frac{n^2 \pi^2 Dt}{4R^2}\right)$$

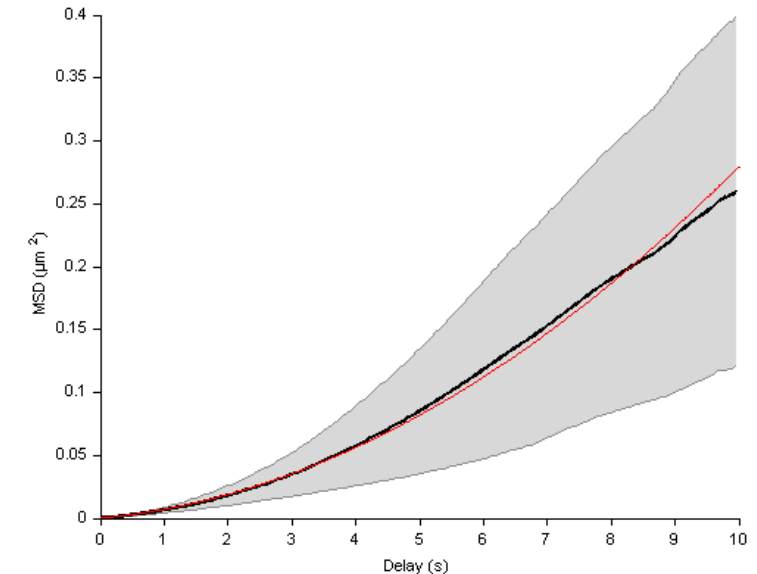
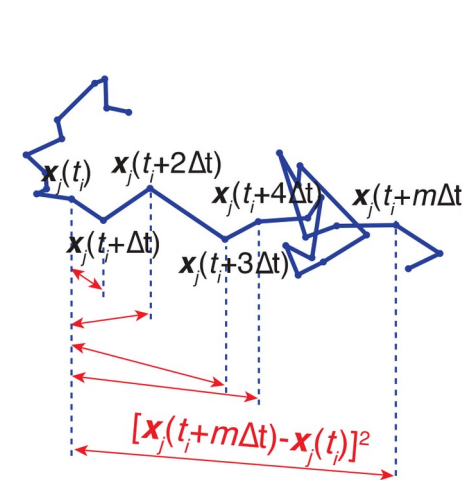
MSD analysis - care should be taken

Localization error



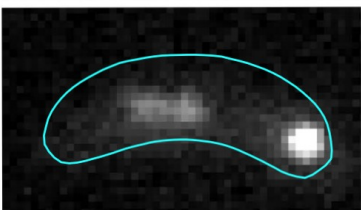
Localization error adds a constant term to the MSD, which, if not properly accounted for, can limit or bias the exact quantification and interpretation of the MSD data

Large lag time



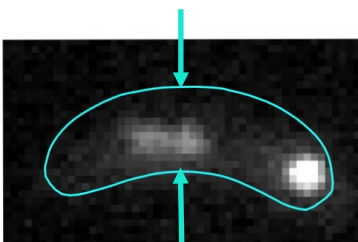
Because of the finite trajectory length, at large time lags their average is computed over a smaller number of data, resulting in more scattered data points and less statistical significance

Motion blurring



While such motion blur contains potentially useful information (Elf et al., 2007), it has some detrimental consequences: a decrease of the SNR and the ineffectiveness of traditional Gaussian fit localization algorithms.

Finite volume



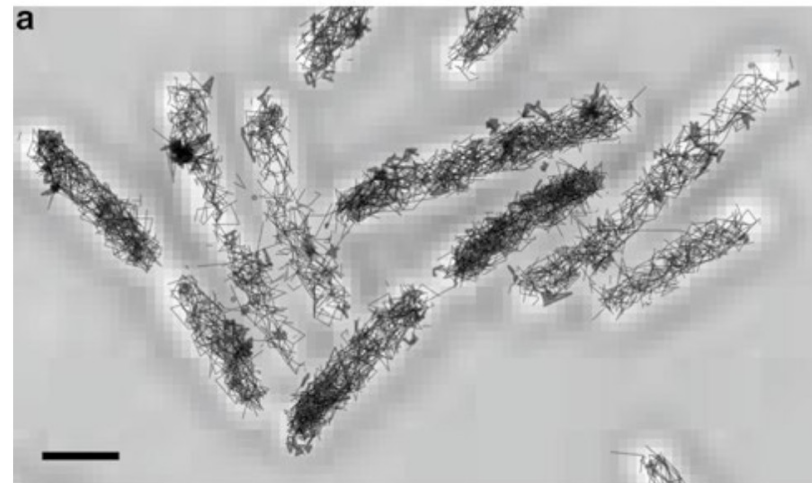
Boundaries will bias:

- particle direction
- particle behaviour (confined)

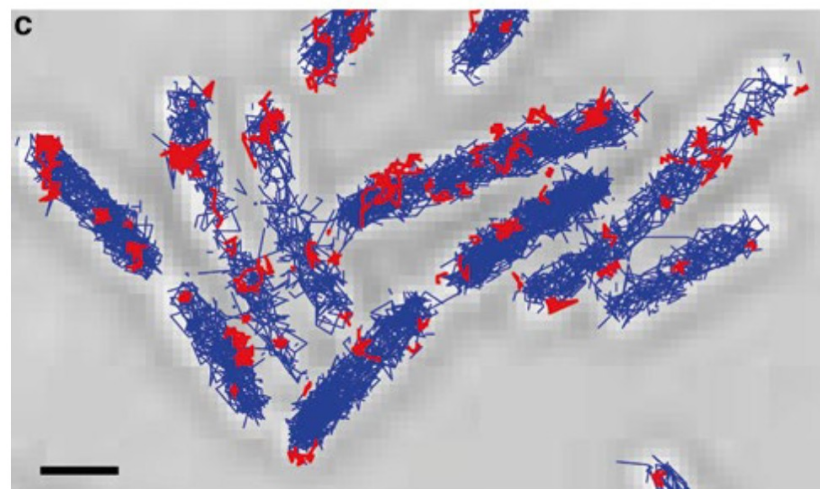
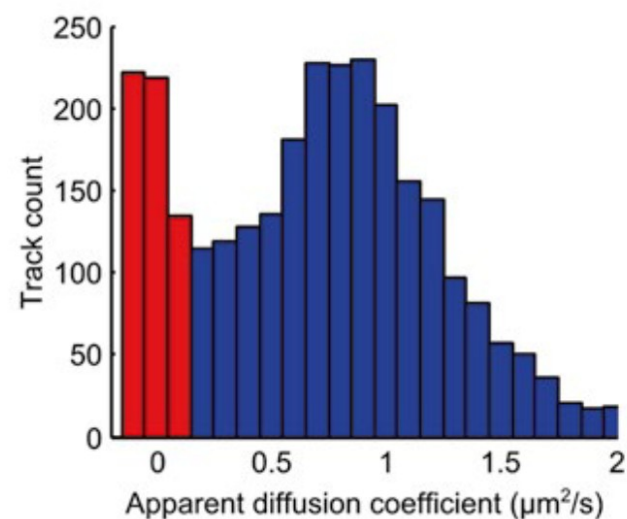
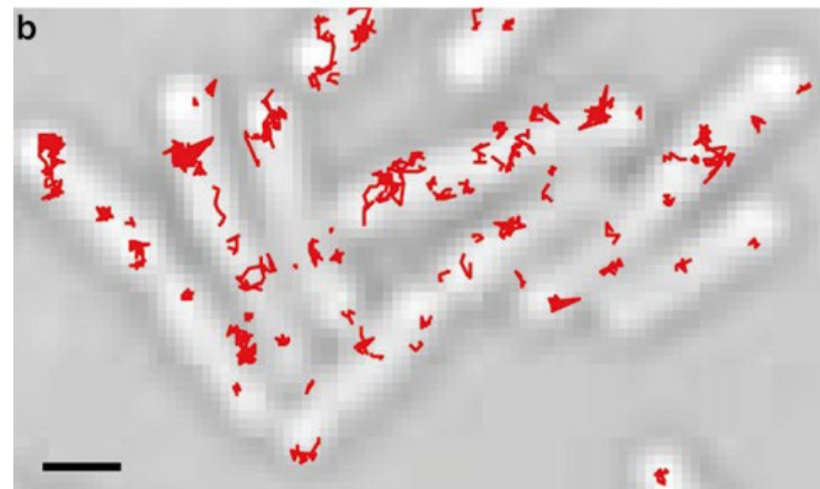
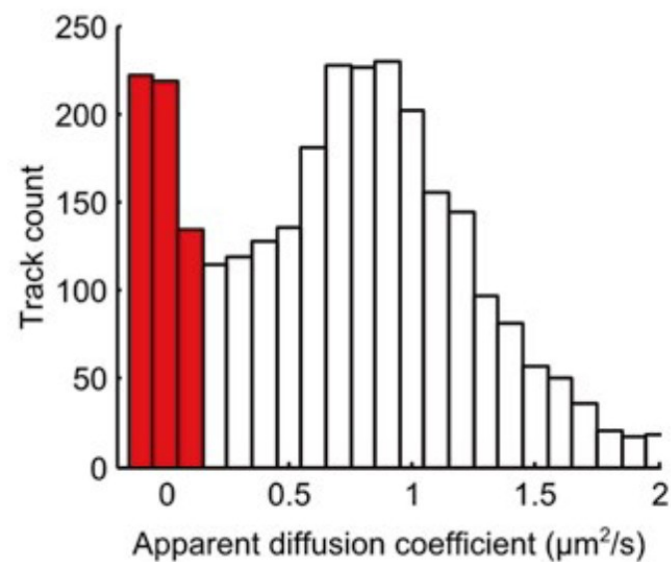
Trajectories too short

Not enough statistics : even a population with 100% brownian particles will be analysed as a fraction of brownian, directed and confined.

Particle localisation - case study - part 3/3



**Trajectories too short
to extract directly
particles behaviour**



More info from SPT

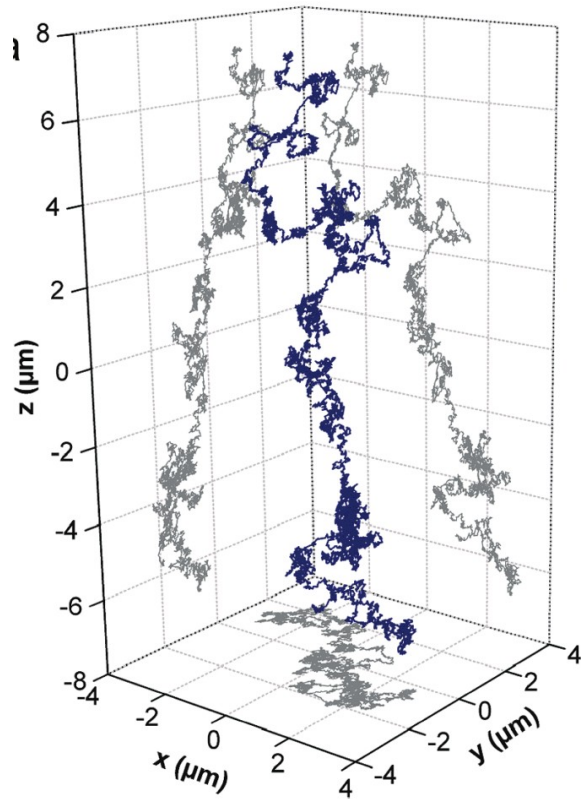
More questions with single particle tracking :

- Where these particles are going and where they are coming from.
- Do they seem to cluster somewhere?
- Do you see any pattern about their orientation relative to the cell?
- Is there net flow inwards or outwards? Vorticity?
- What is the structure of these cellular highways?
- How fast are particles moving?
- In which directions?
- How consistent are particles' velocities in a particular region?
- What does this say about how they are propelled?
- Do particles move consistently or do they change direction and speed?
- Do particles move uni-directionally or do they cross each other?
- Generally, what is the velocity distribution of directed particles, and what are the variables of this distribution?
- ...

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Why tracking in 3D?



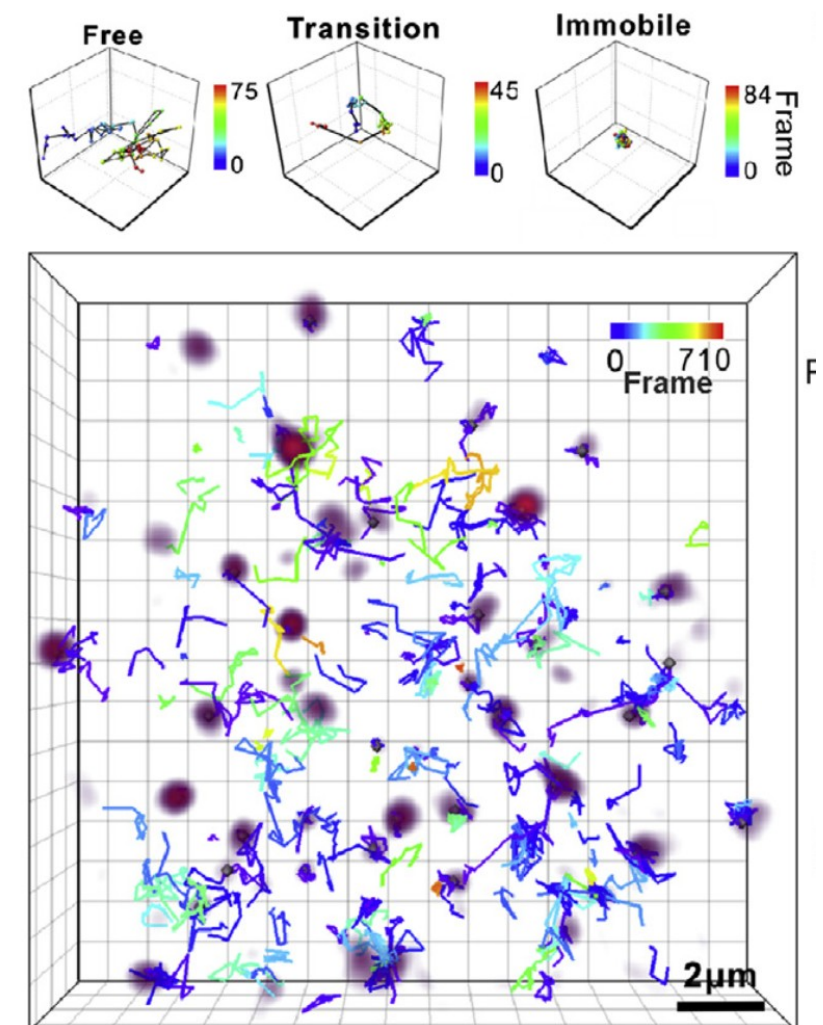
Tracking 3D motion in 2D :

- part of the information contained in the real trajectory is missing in the analysis of the 2D projected trajectory
- misevaluation of the motion parameters such as the diffusion coefficient or the velocity.

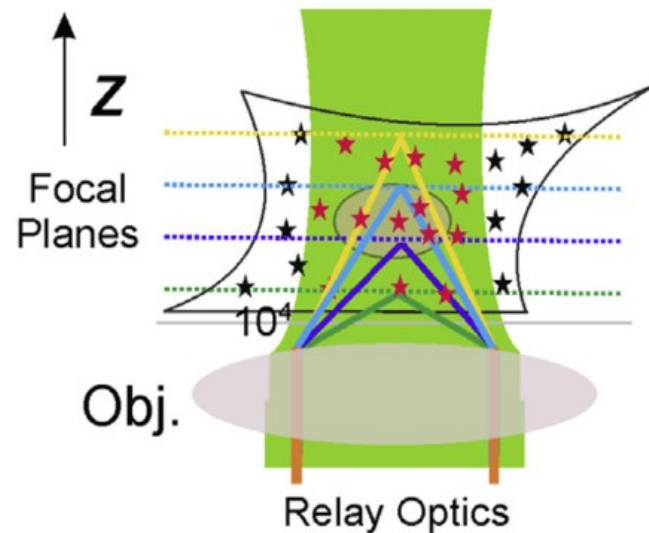
Life occurs in three dimensions

- Extend SPT experiments to
- intracellular / nuclear compartments,
 - multicellular systems
 - ...

More challenging



Z scanning



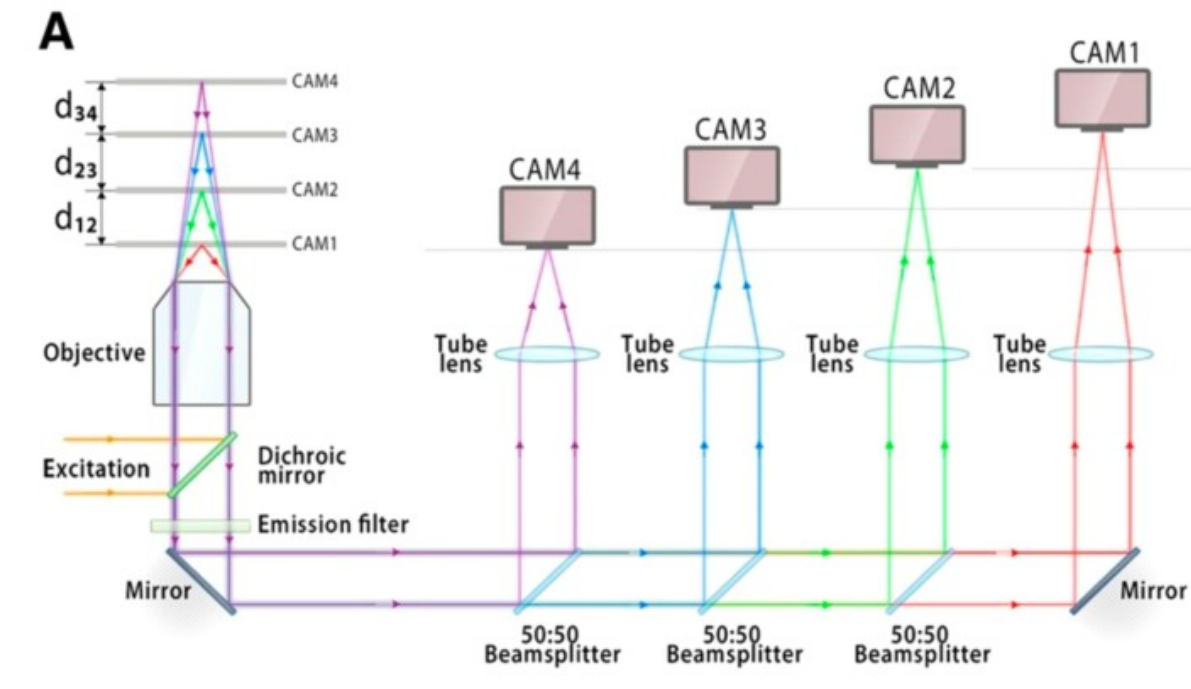
Principle : scanning over different z depths and recording the corresponding images

Pros :
- Accessibility

Cons :
- Photobleaching
- Slow

restricted to extremely slow dynamic processes

Bi & Multi-Plane Microscopy



Principle : record simultaneously different focal planes on several detectors

Pros :

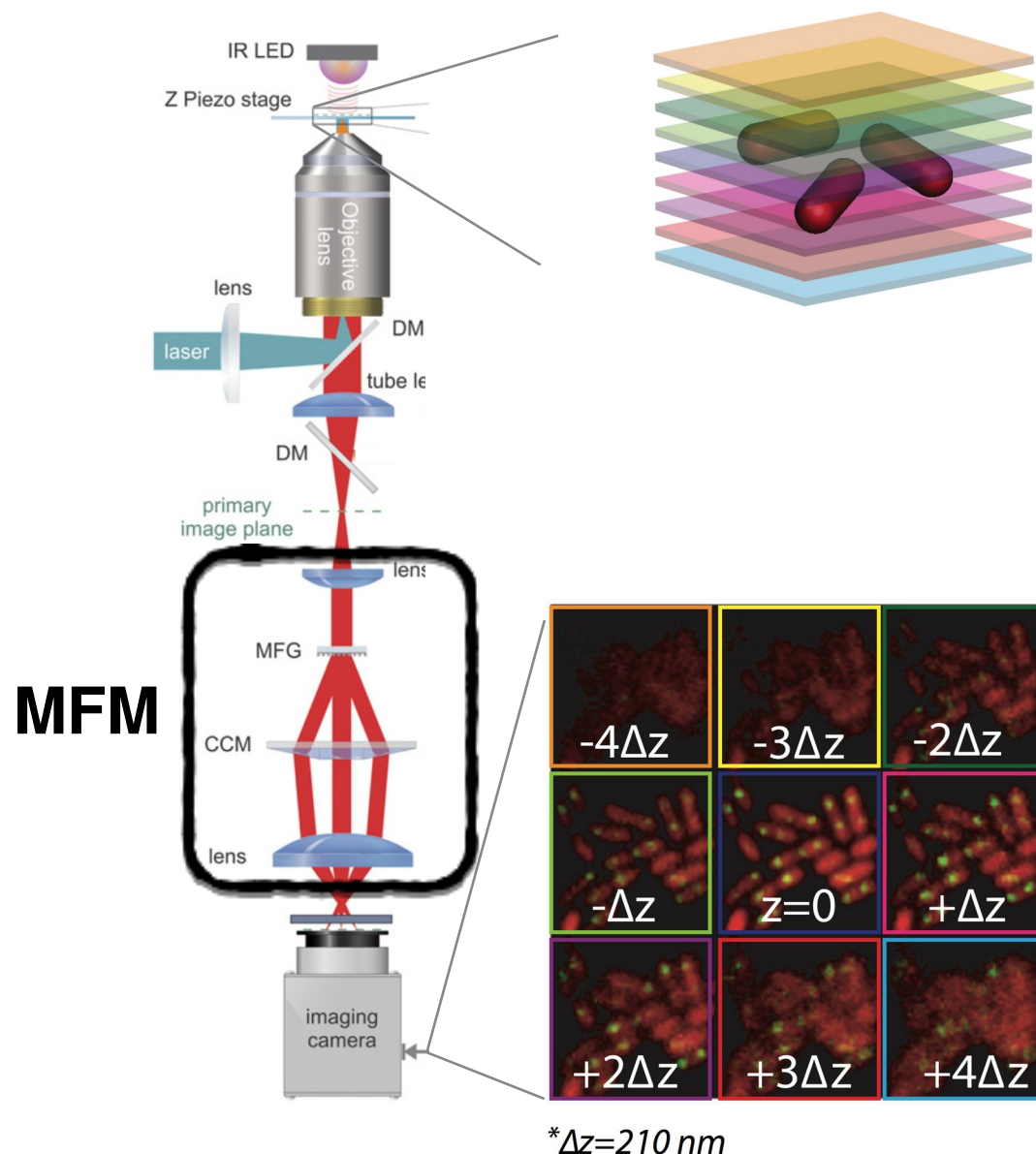
- simultaneous images at different focal planes
- better temporal resolution
- large depth of field

Cons:

- Split photon flux
- Expensive

Multi Focus Microscopy

Multi-Focus Microscope:



Principle : record simultaneously different focal planes on a single detector with diffractive optics

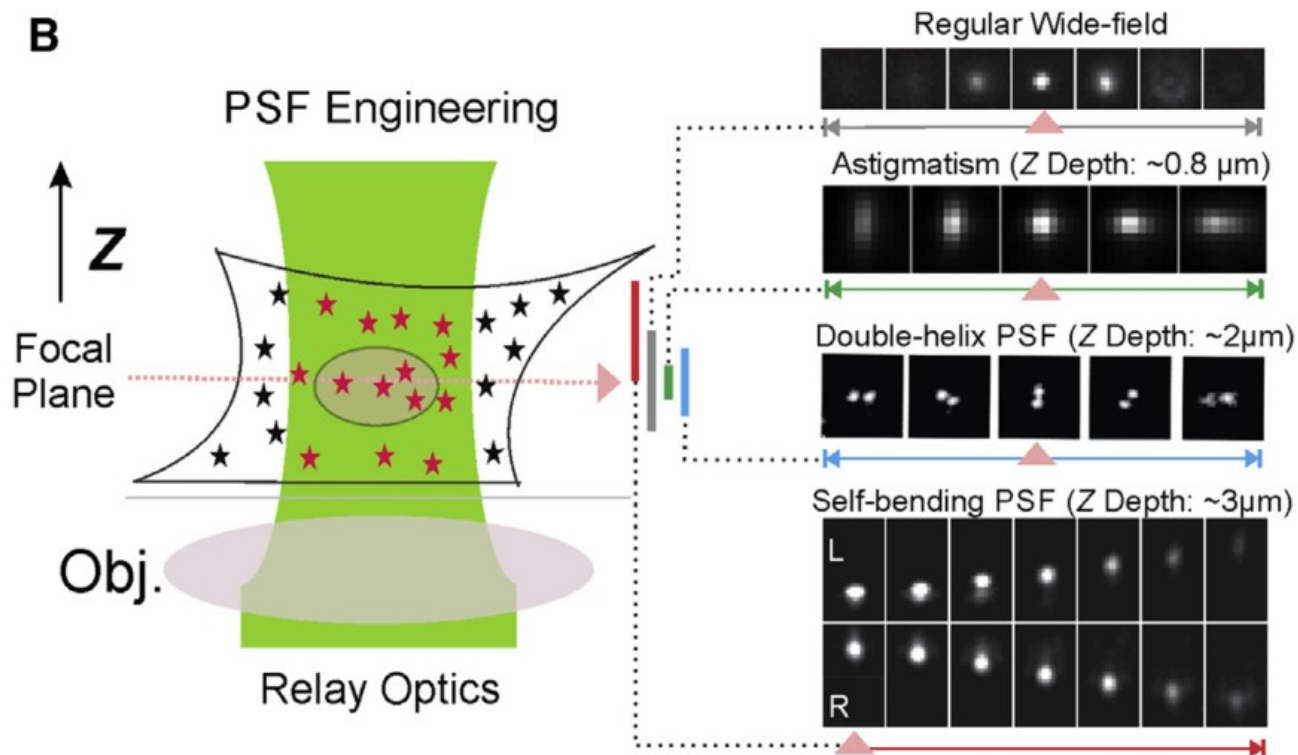
Pros:

- simultaneous images at different focal planes (>9)
- great temporal resolution
- very large depth of field ($>4\mu\text{m}$)
- Photons split based on focal plane

Cons:

- Split field of view

PSF engineering



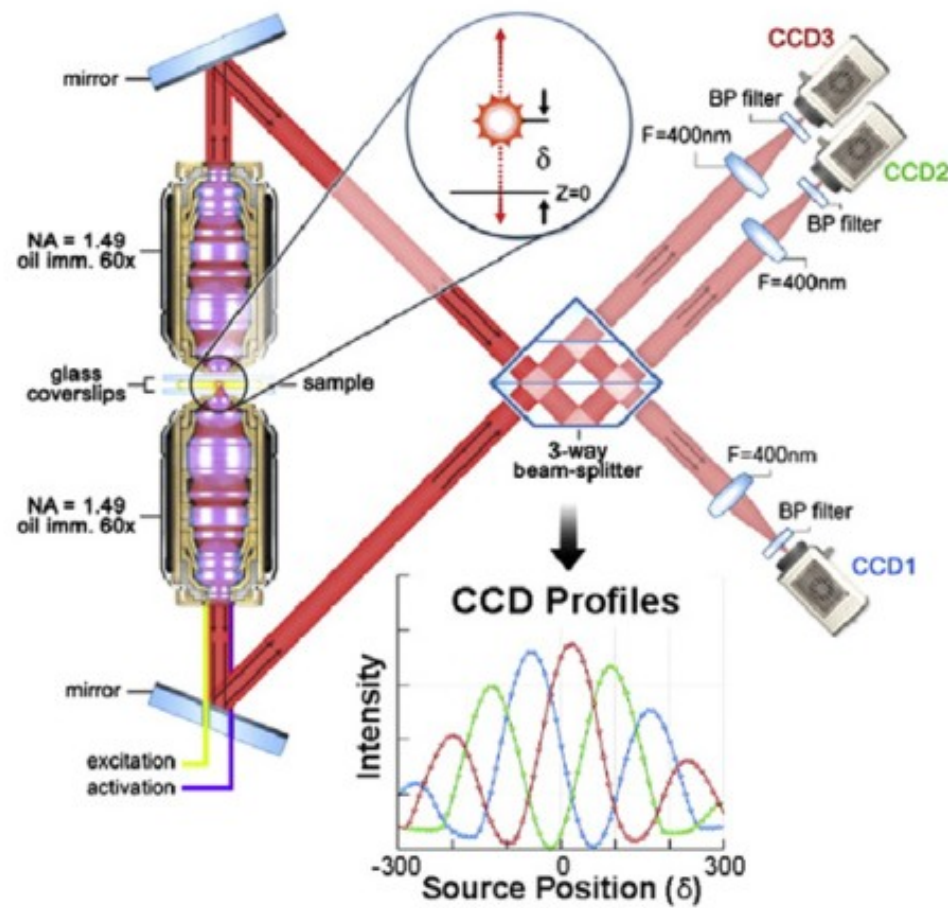
Principle: takes advantage of the size and shape of the intensity spot (the PSF of the microscope) when is de-focused to infer on the z-position of the particle (figure

Pros:

Cons:

- Specific localisation algorithms
- lowered emitter density per image (overlapping of PSFs from different depths)
- Phase masks are color-specific

Interferometric PALM



Pros:

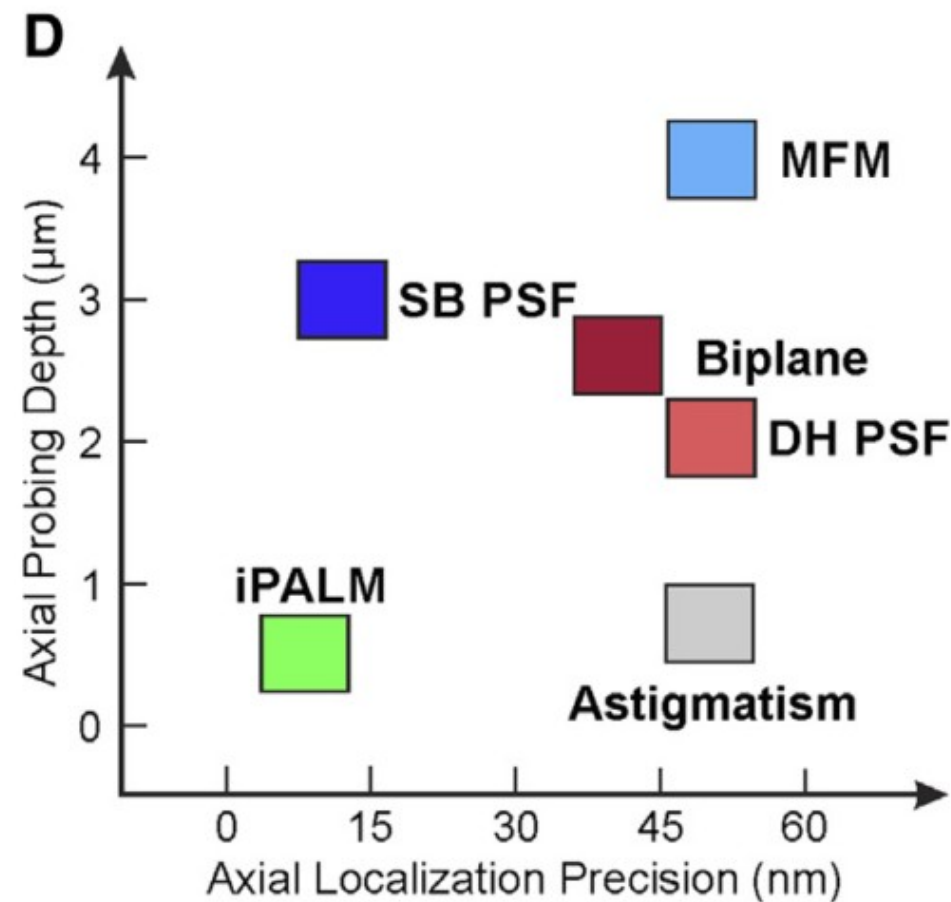
- Resolution

-

Cons:

- Complexity
- Limited depth of field

3D techniques DOP and Z precision



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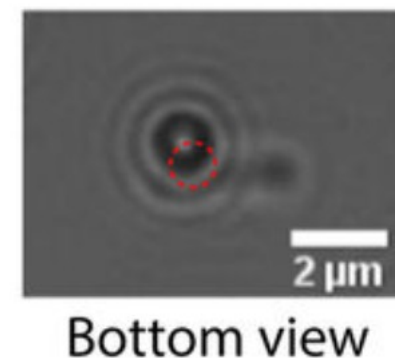
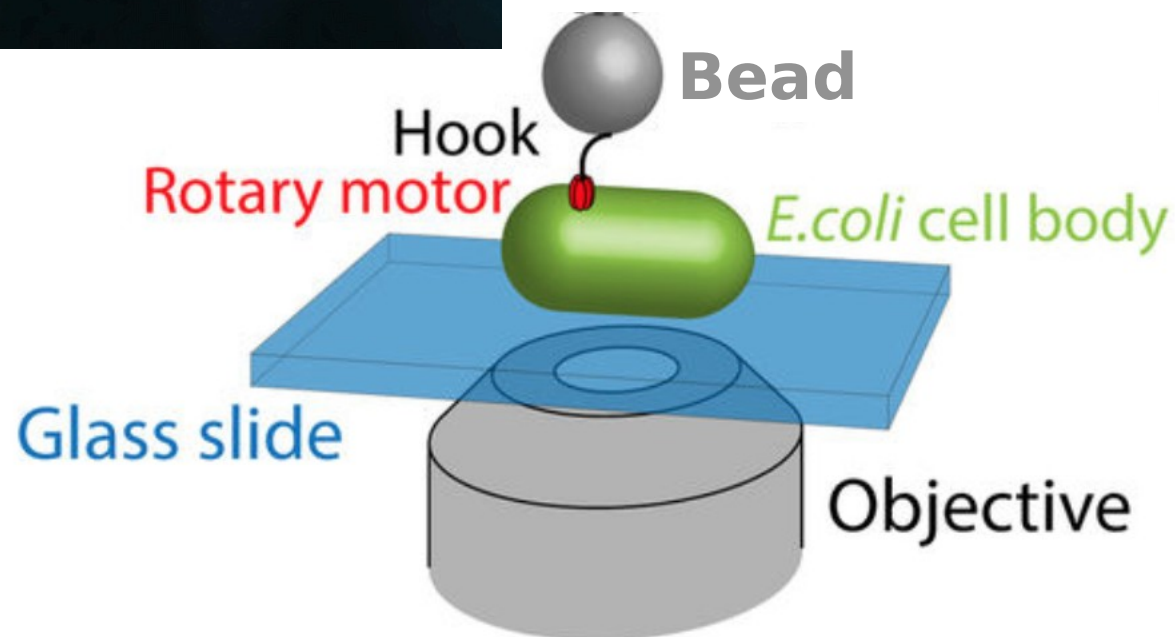
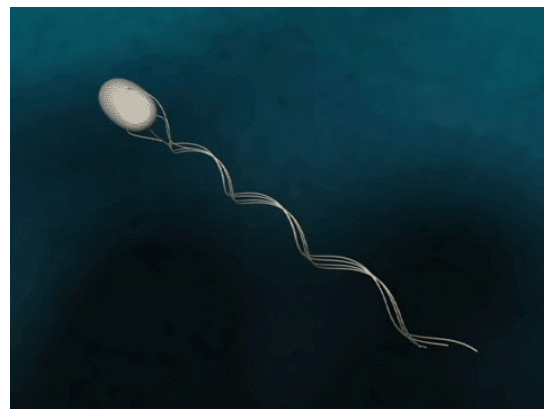
Single Particle Tracking at CBS



Flagellar motor

Single Particle tracking Optical and Magnetic Tweezers

Torque generation and angular dynamics of the bacterial flagellar motor



- > Torque
- > Angular speed

Membrane trafficking



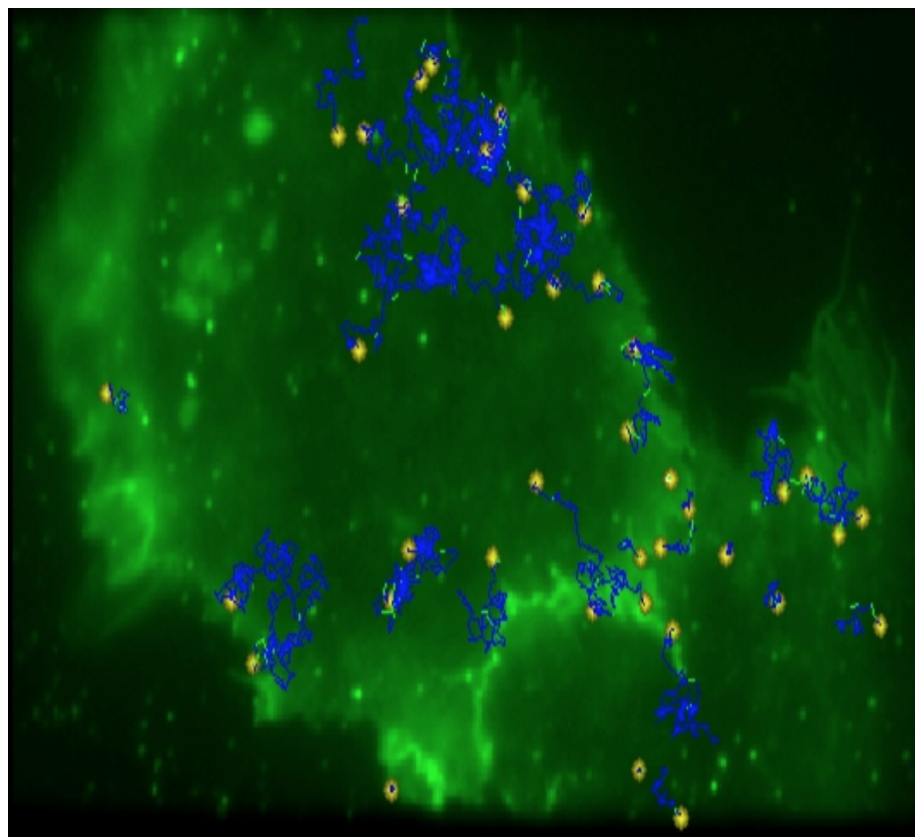
P. Milhiet
E. Margeat

Dual-color Single Particle tracking

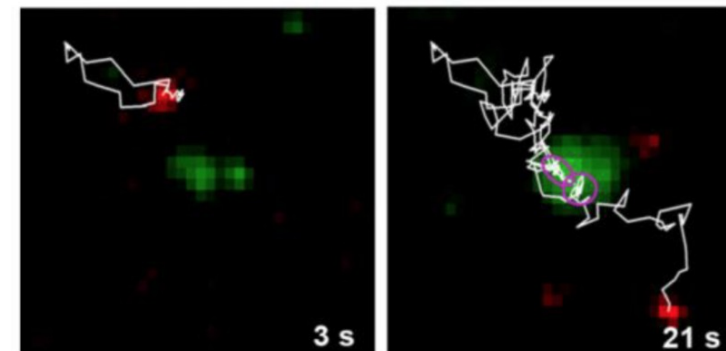
Atomic Force Microscopy

Super-Resolution

Structure and dynamics of membrane assemblies



transmembrane proteins that form supramolecular assemblies organized in microdomains



Dosset et al, BMC bioinformatics, 2016

Potel et al, Cell Microbiol, 2013

Chromosomal loci & molecular motors



M. Nollmann

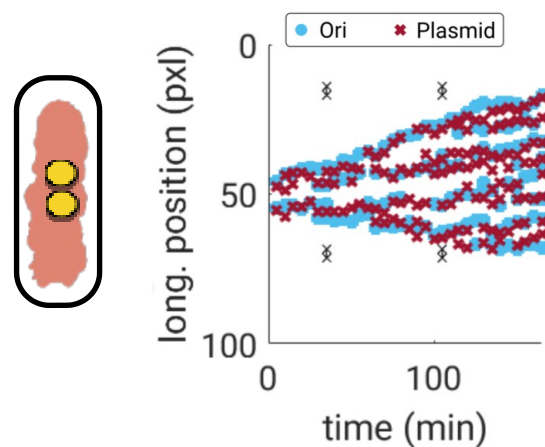
spt-PALM

Dual color tracking

Correlative super-resolution & tracking

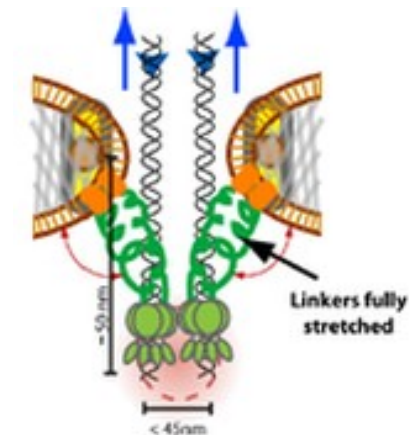
3D tracking

Chromosome organisation and segregation

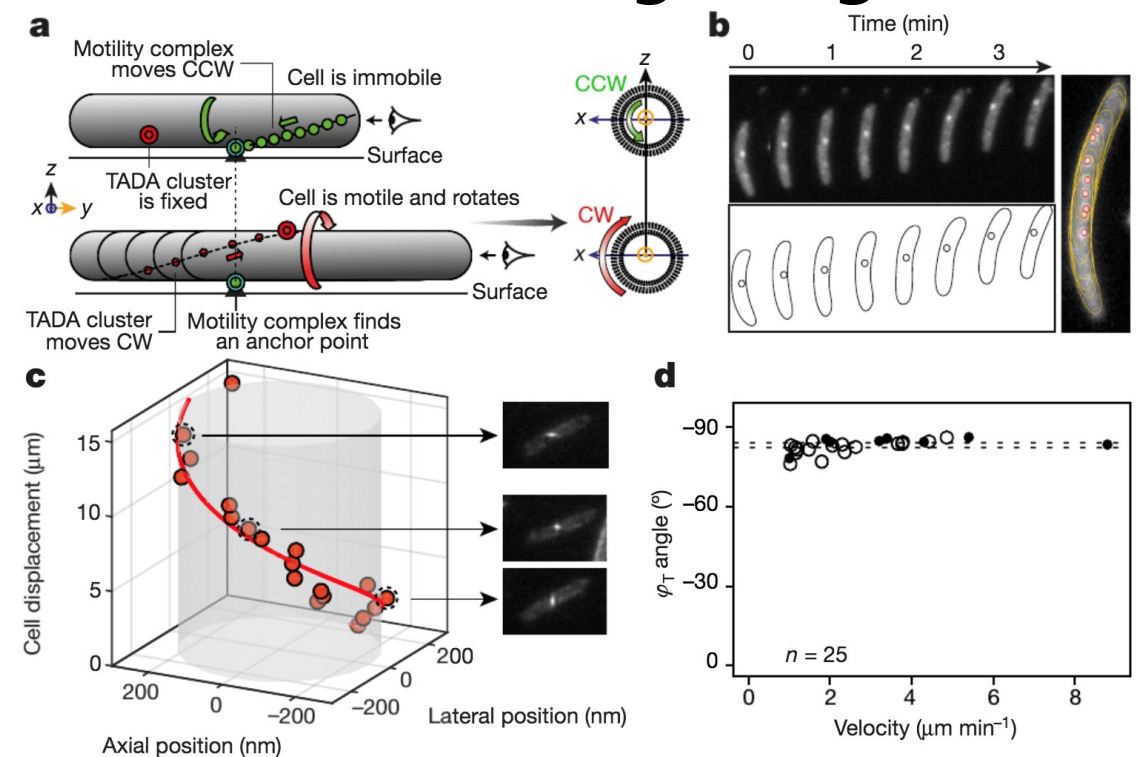


Le Gall et al, Nature Communication, 2016

DNA molecular motors



Bacterial gliding



Faure, Fiche et al, Nature, 2016

References

Shen et al, Single Particle Tracking: From Theory to Biophysical Applications, Chemical Reviews, 2017

Manzo et al, A review of progress in single particle tracking: from methods to biophysical insights, Reports on Progress in Physics, 2015